

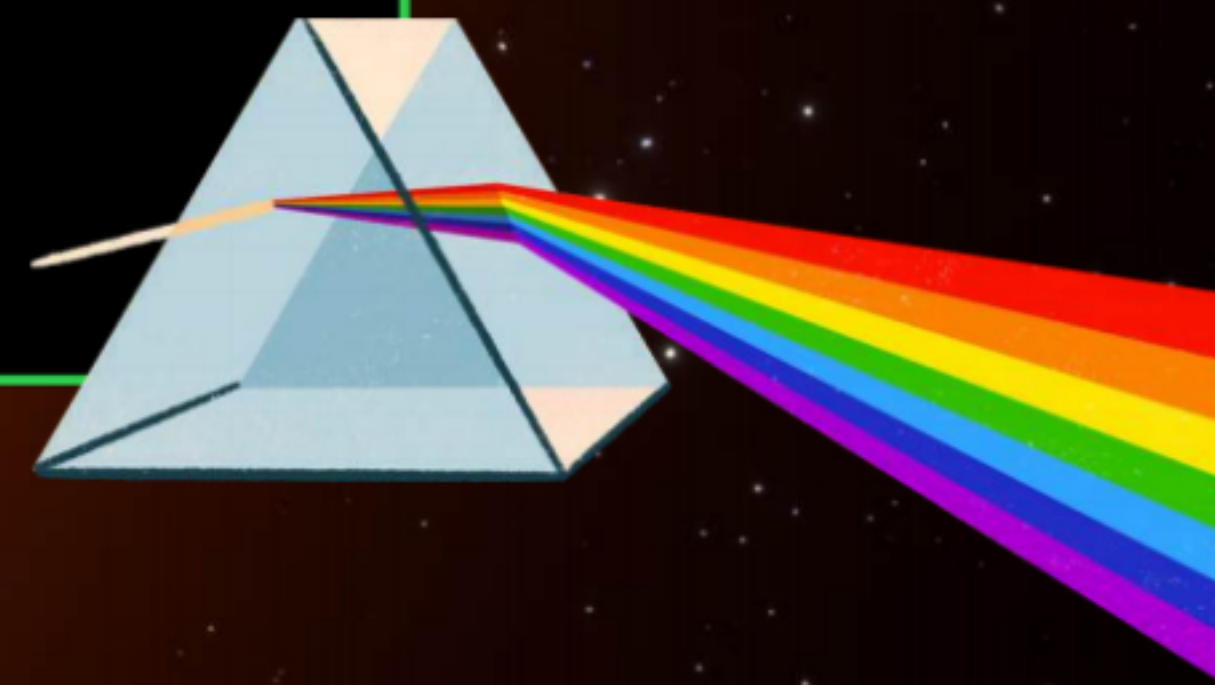
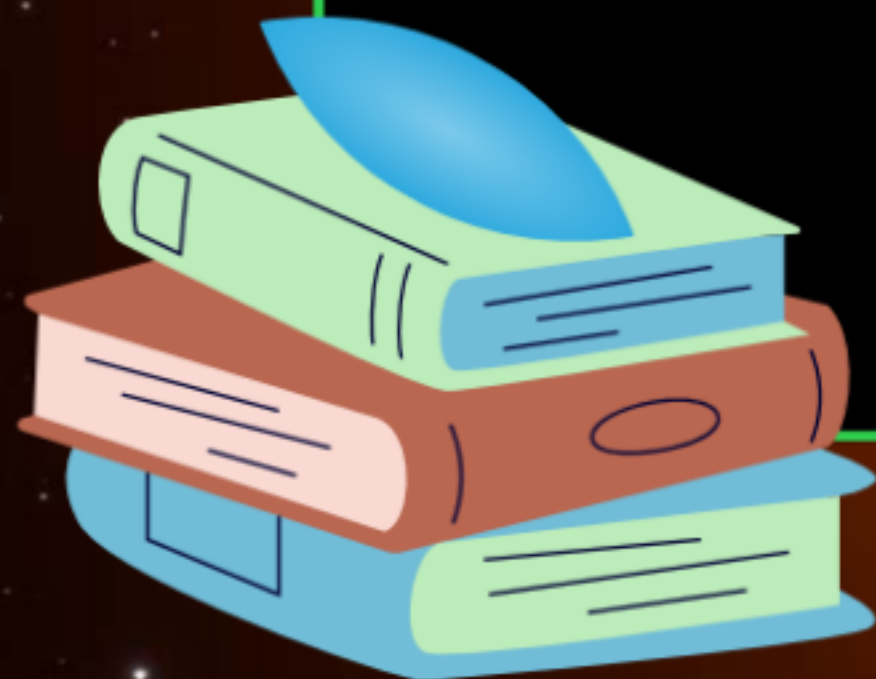
L6

आरंभ
BATCH

CLASS X - SCIENCE

LIGHT

PRASHANT KIRAD



Lens

A lens is a transparent material bounded by two surfaces, out of which one or both surfaces are spherical.

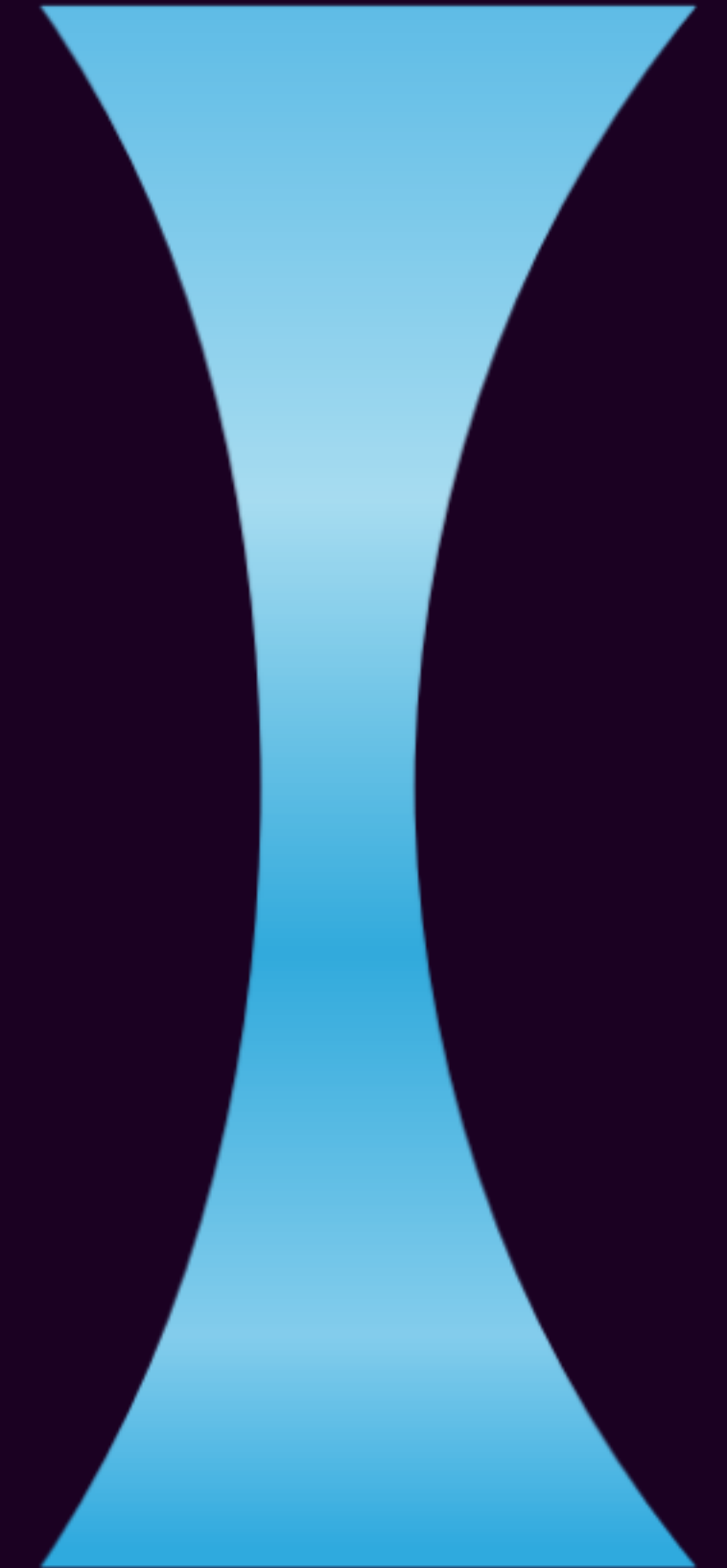
This means a lens must have at least one curved spherical surface.

The other surface may be plane or spherical.

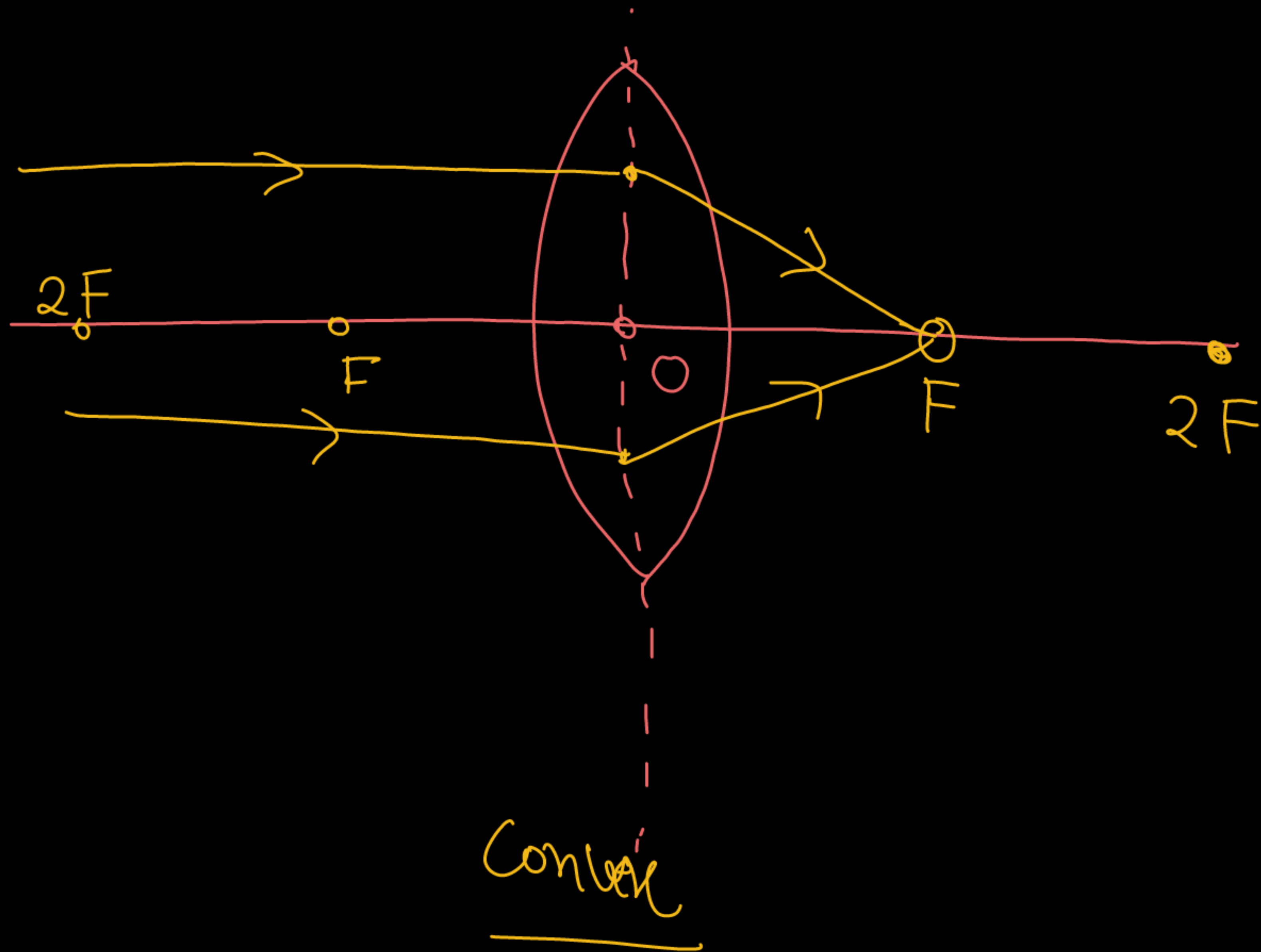
glass

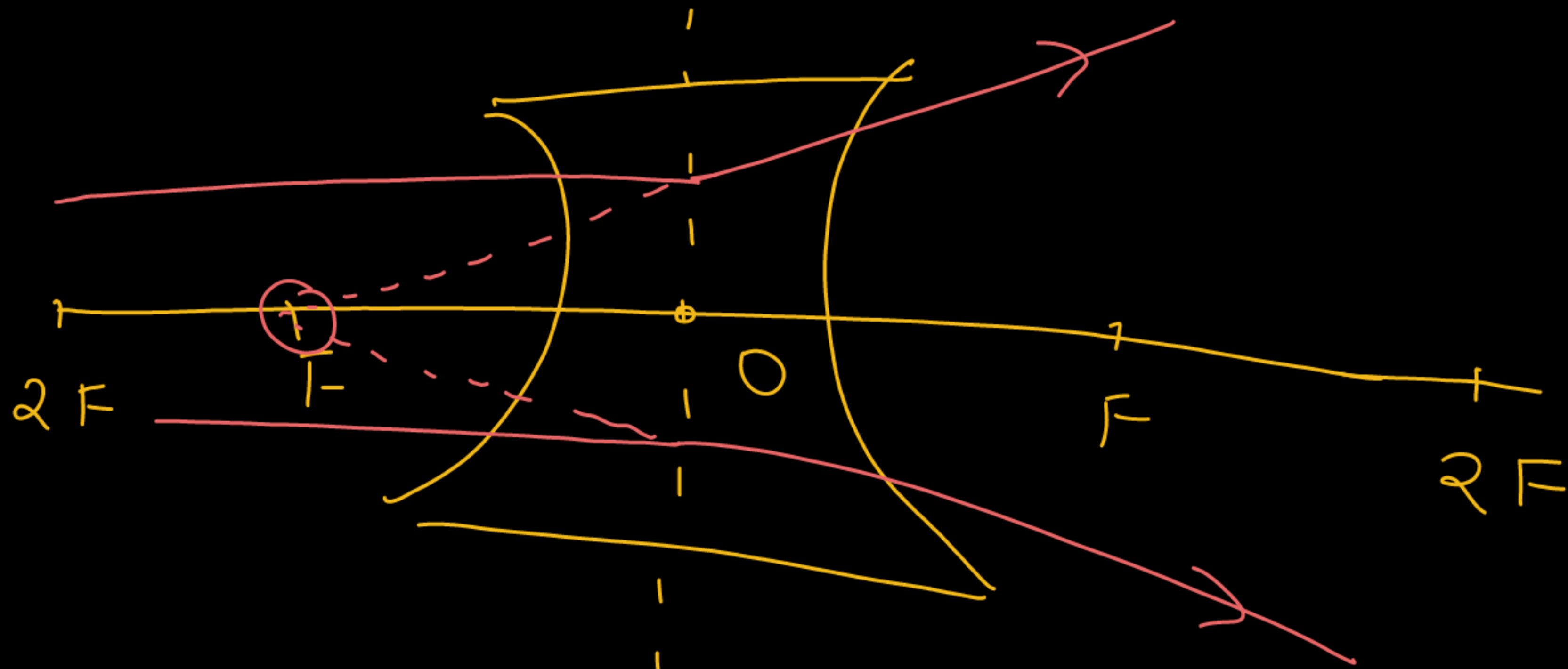


Convex Lens



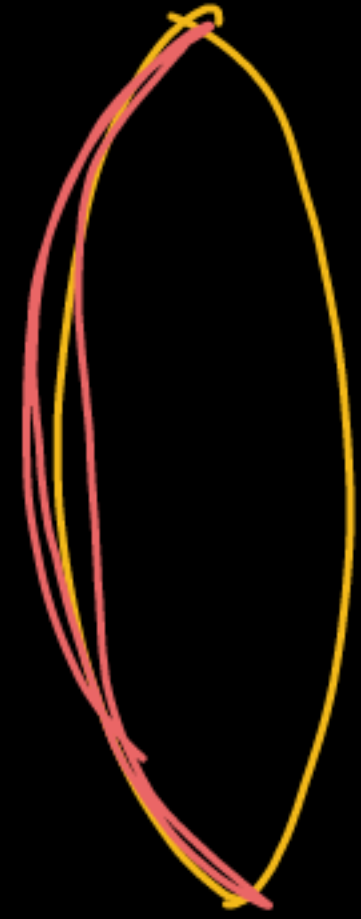
Concave Lens



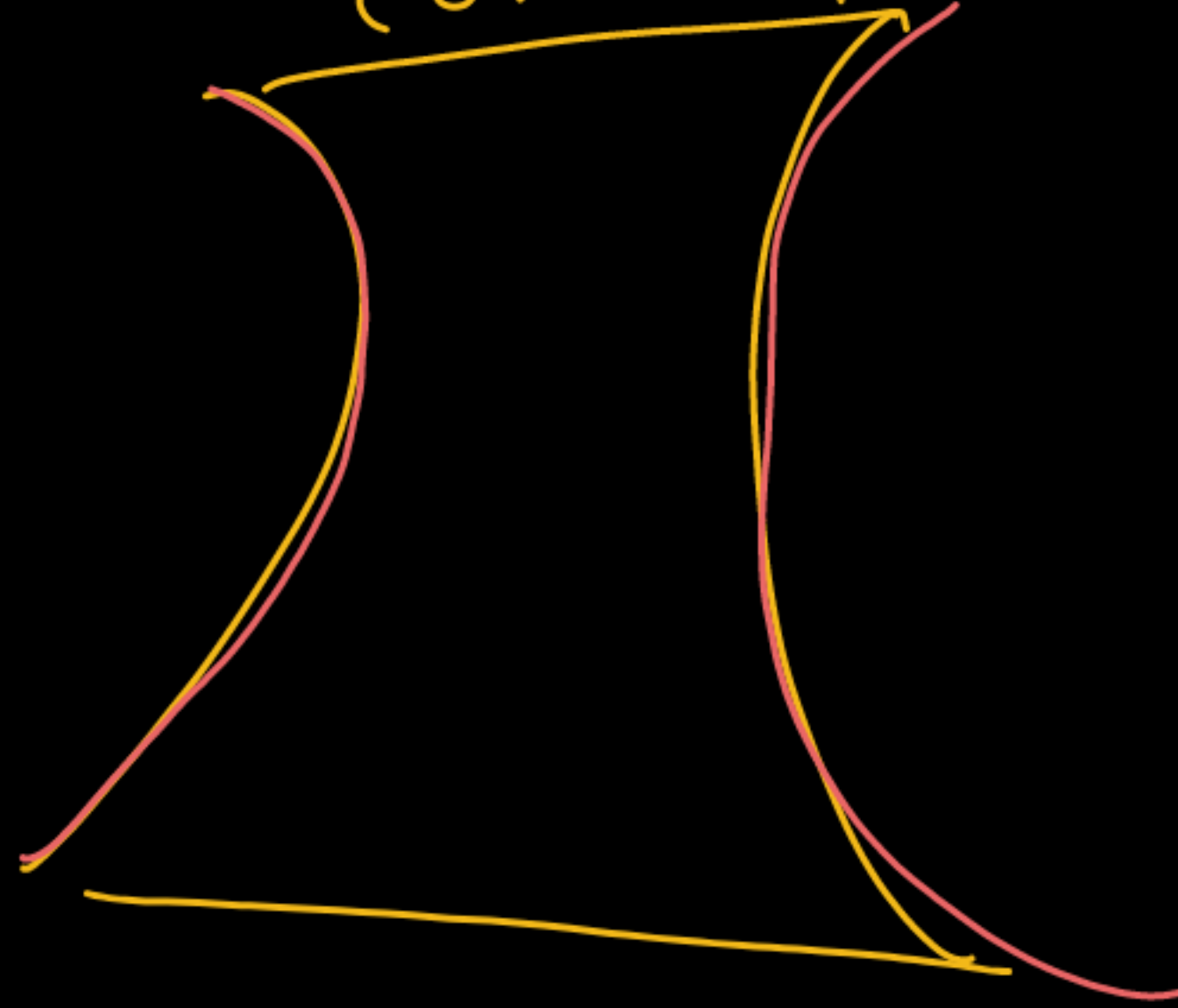


diverging ✓

convex



concave



Convex Lens

A lens having two spherical surfaces bulging outwards is called a double convex lens.

- Convex lens is thicker at the middle and thinner at the edges.
- It converges light rays.
- Therefore, it is also called a **converging lens**.

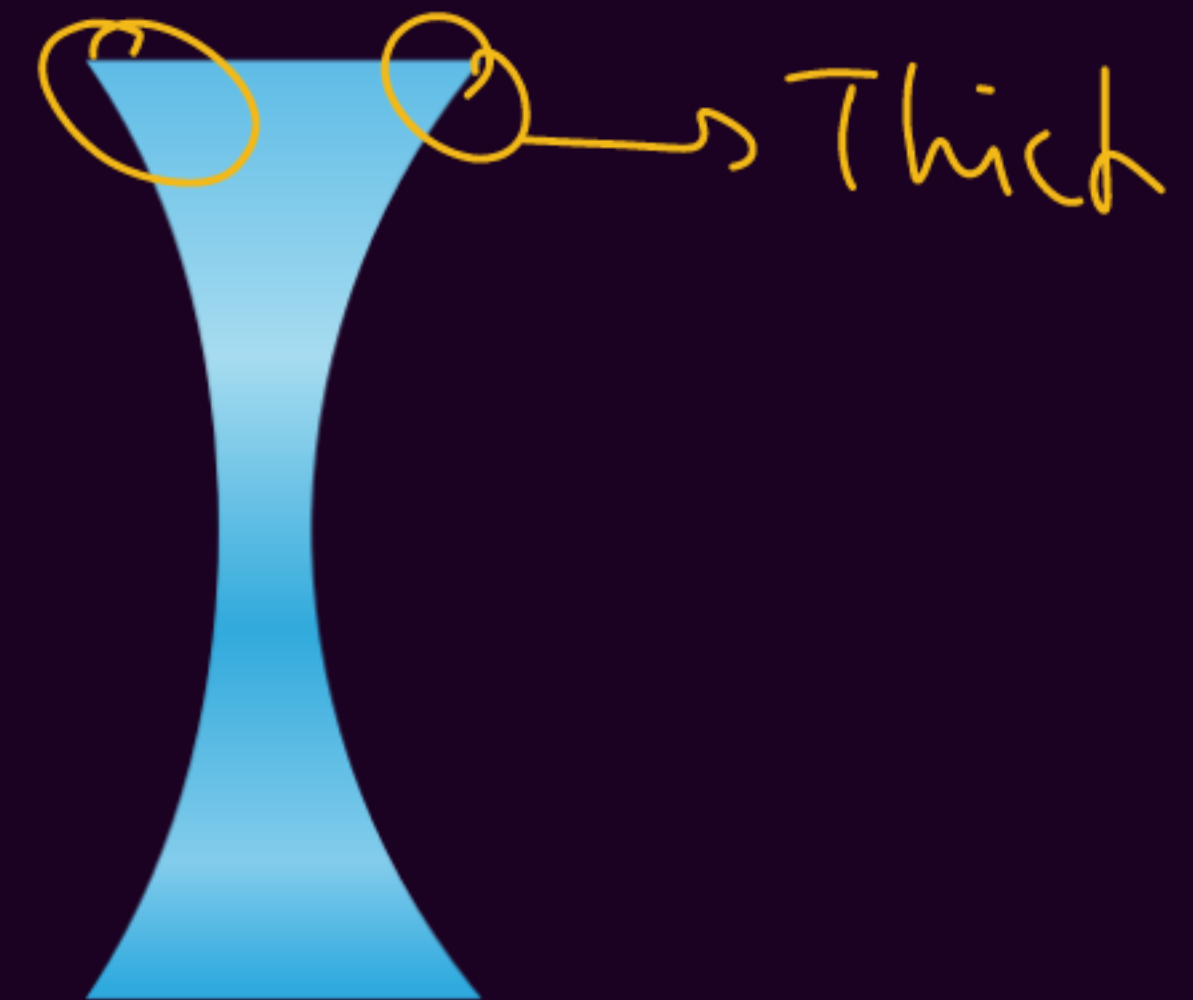


7 June - 14 June

Concave Lens

A lens having two spherical surfaces curved inwards is called a double concave lens.

- Concave lens is thinner at the middle and thicker at the edges.
- It diverges light rays.
- Therefore, it is also called a **diverging lens**.



Important Terms:

✓ Principal Axis:

An imaginary straight line passing through the two centres of curvature of a lens is called the principal axis.

It passes through C_1 , O and C_2 .

Optical Centre:

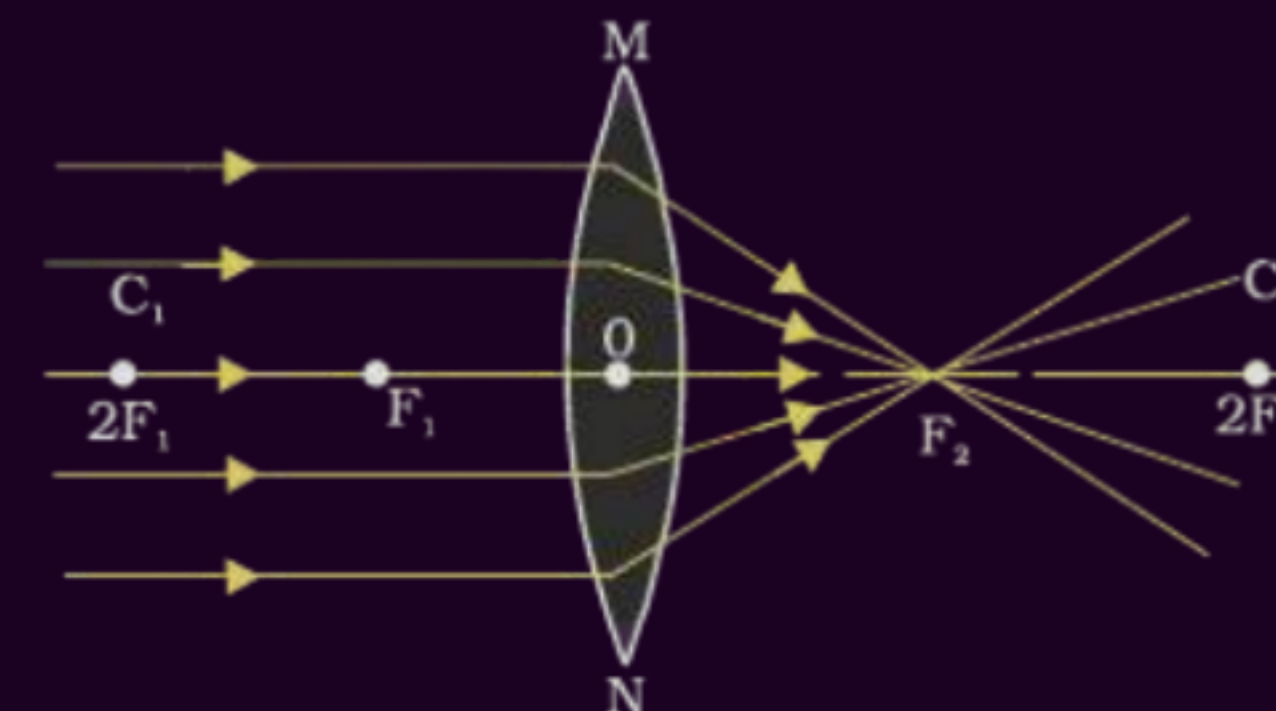
The central point of a lens is called its optical centre.

It is represented by the letter O .

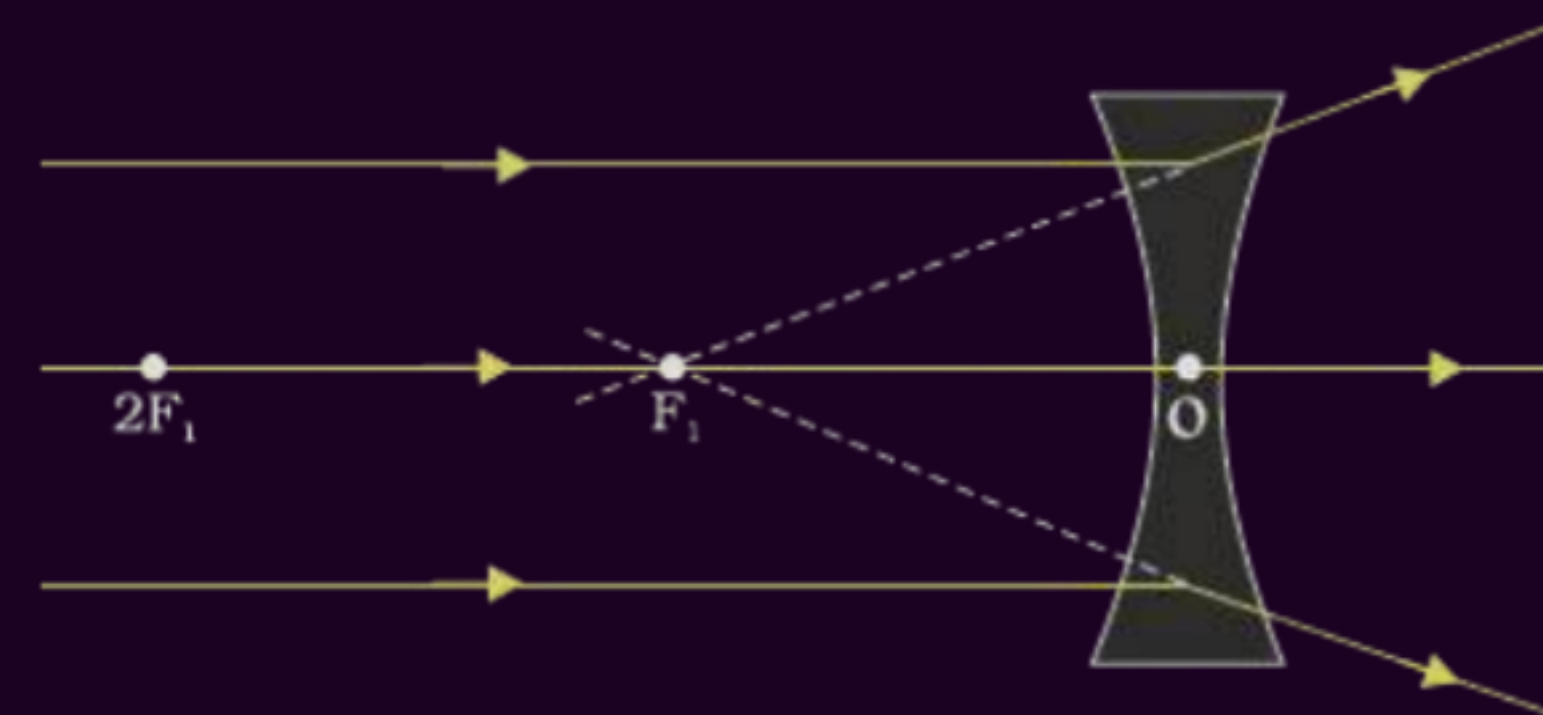
A ray of light passing through the optical centre of a lens goes undeviated.

Aperture of a Lens:

The effective diameter of the circular outline of a spherical lens is called its aperture.



(a) Converging action of convex lens



(b) Diverging action of concave lens

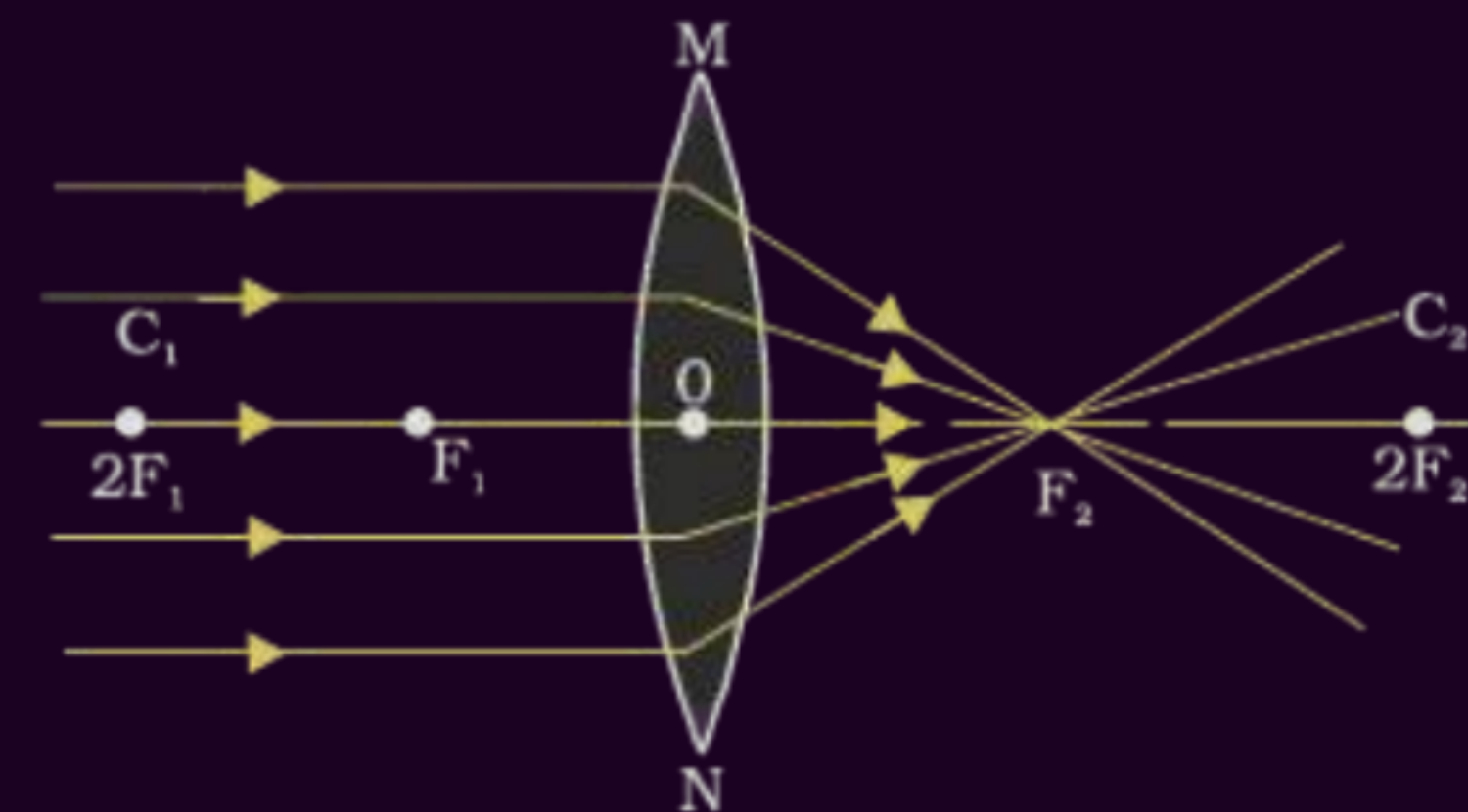
A thin lens is a lens whose aperture is much smaller than its radius of curvature.

Principal Focus of Convex Lens

The principal focus of a convex lens is the point on the principal axis where rays of light parallel to the principal axis meet after refraction.

Convex lens has a real focus because light rays actually meet there.

F₁ and F₂ are the foci of the convex lens.



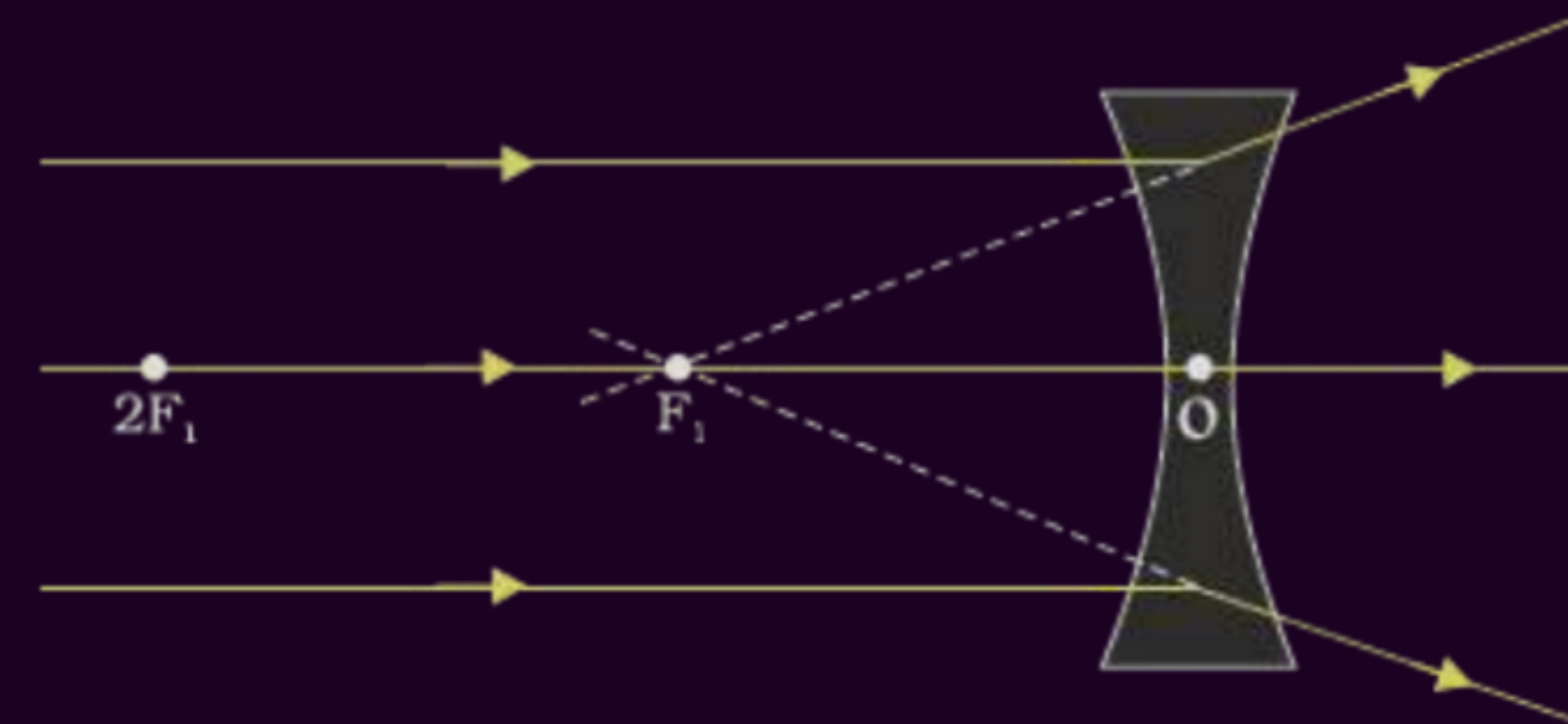
(a) Converging action of convex lens

Principal Focus of Concave Lens

The principal focus of a concave lens is the point on the principal axis from which rays of light parallel to the principal axis appear to diverge after refraction.

Concave lens has a virtual focus because light rays do not actually meet there.

They are represented as F₁ and F₂.

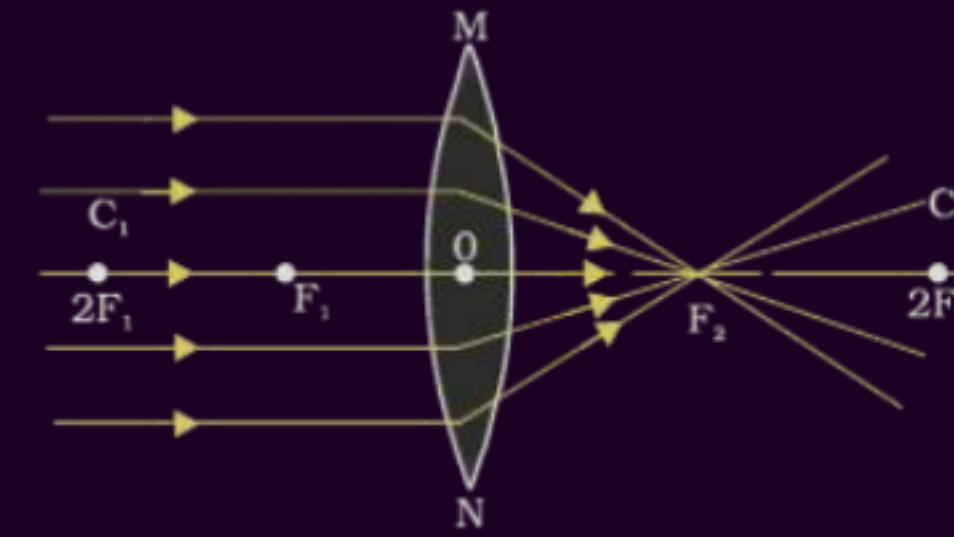


(b) Diverging action of concave lens

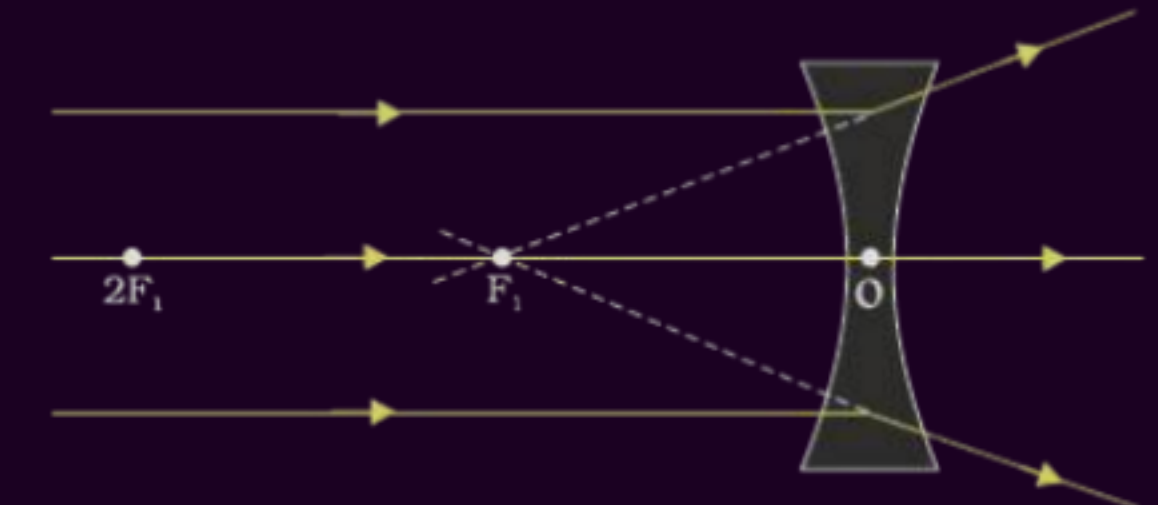
Focal Length of a Lens

The distance between the optical centre and the principal focus of a lens is called its focal length. It is represented by the letter f .

Focal length = distance between O and F



(a) Converging action of convex lens



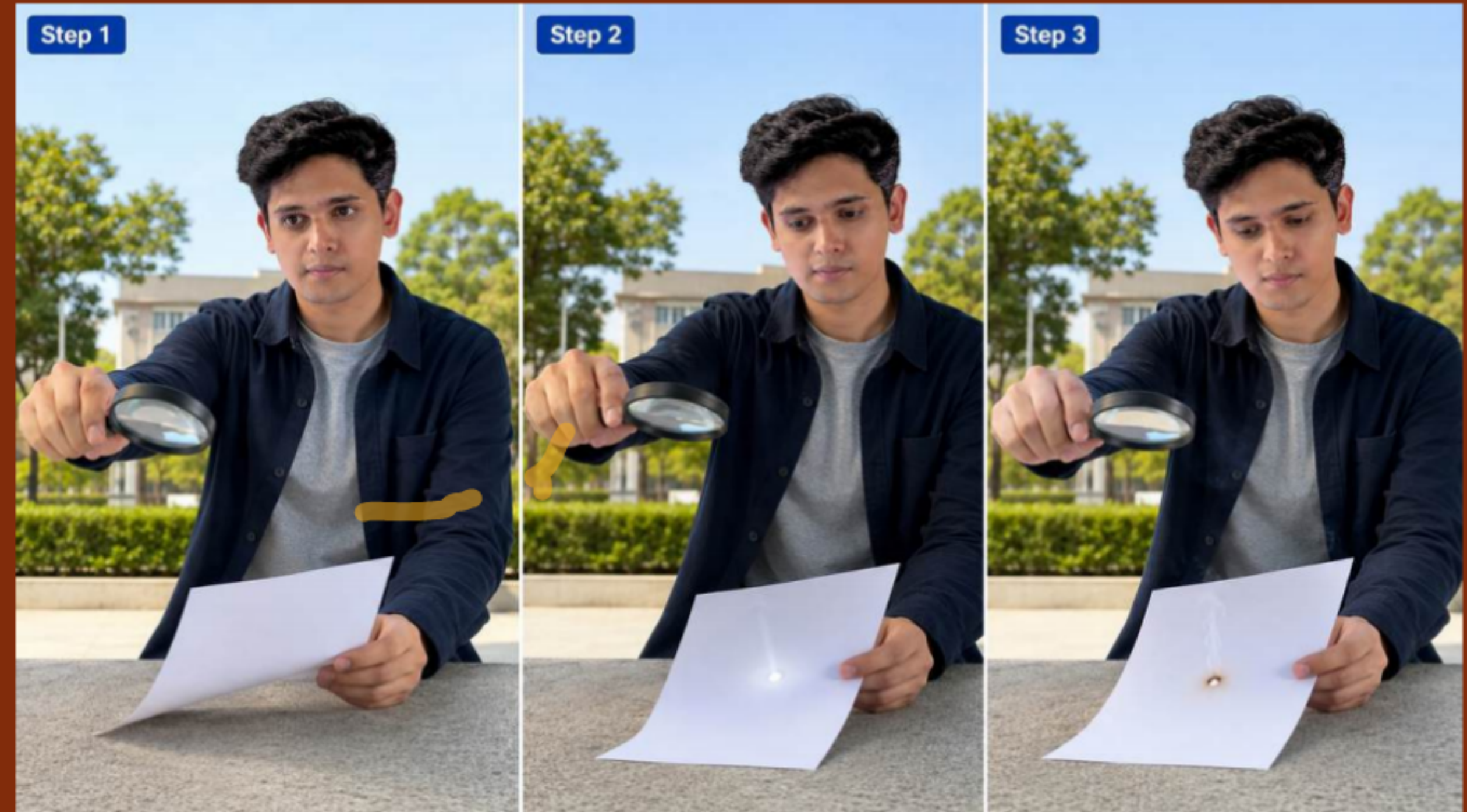
(b) Diverging action of concave lens

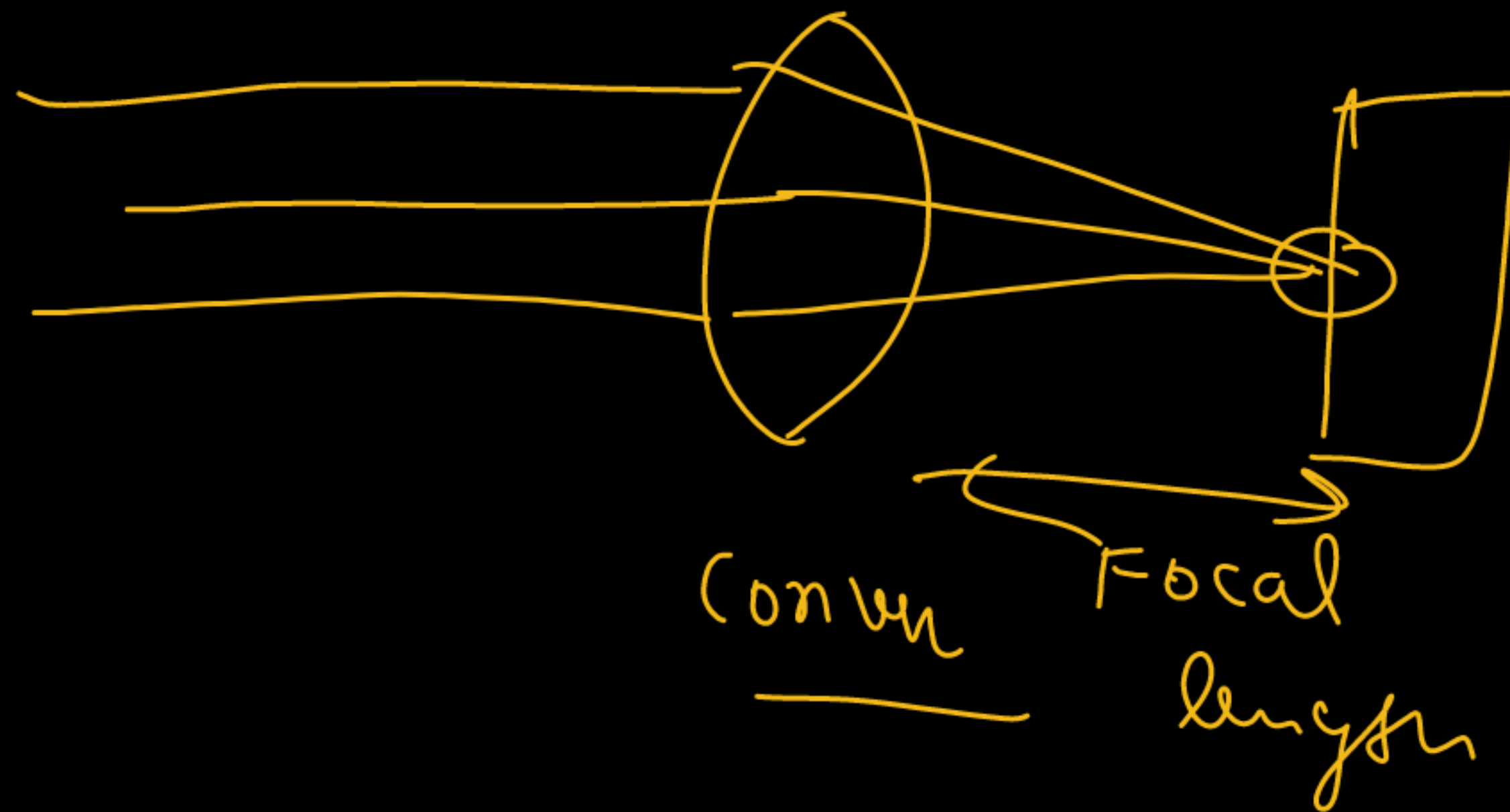
Activity 9.11

CAUTION: Do not look at the Sun directly or through a lens while doing this Activity or otherwise. You may damage your eyes if you do so.

- Hold a convex lens in your hand. Direct it towards the Sun.
- Focus the light from the Sun on a sheet of paper. Obtain a sharp bright image of the Sun.
- Hold the paper and the lens in the same position for a while. Keep observing the paper. What happened? Why? Recall your experience in Activity 9.2.

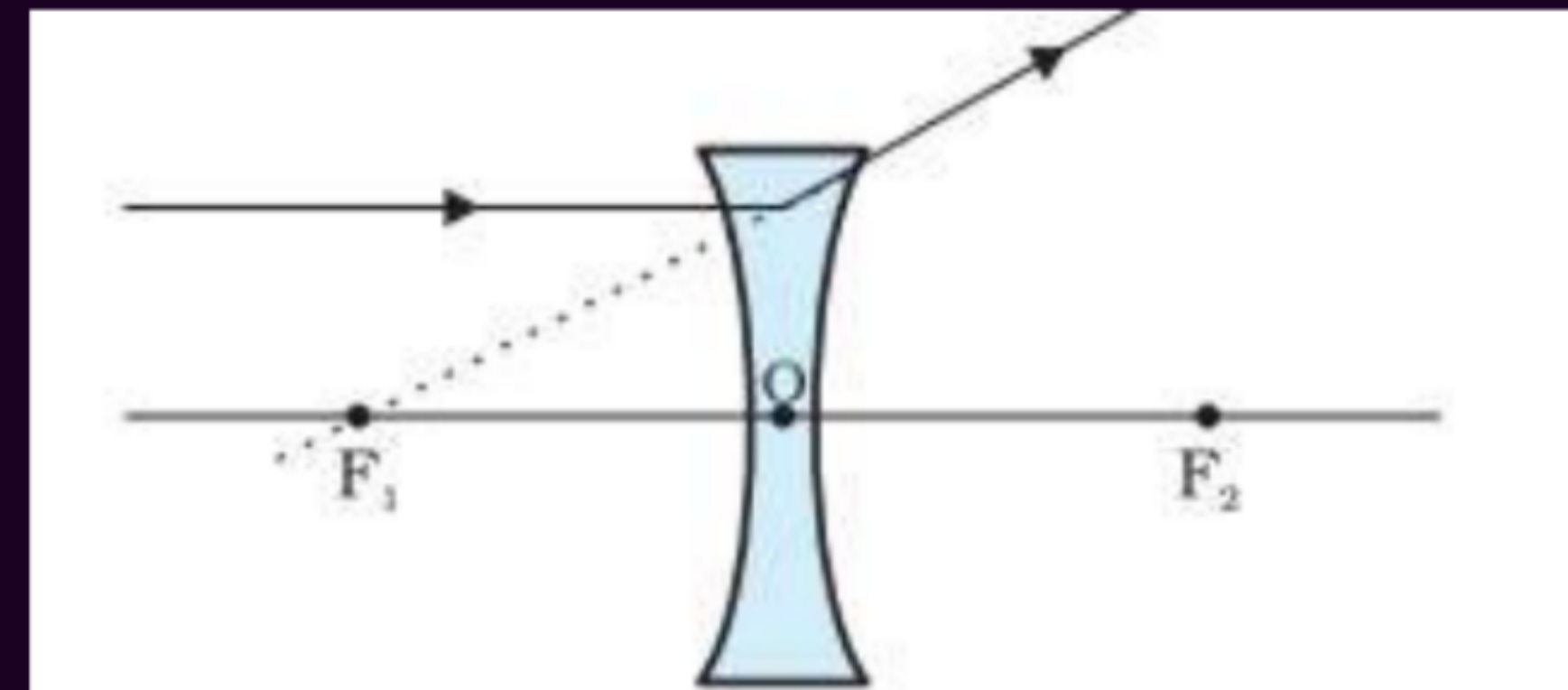
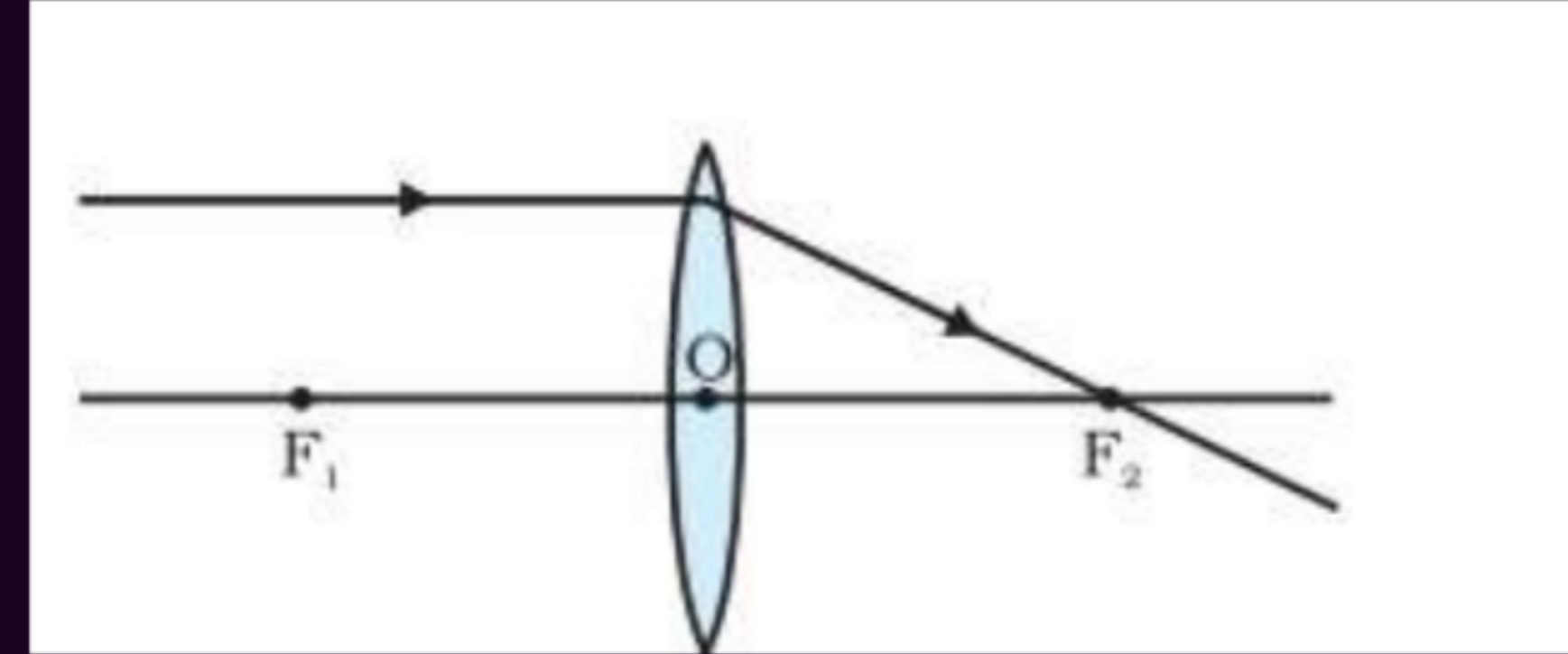
A convex lens converges parallel rays of sunlight at its focus, where heat becomes concentrated. This proves that a convex lens is a converging lens and the bright spot formed is the real image of the Sun.





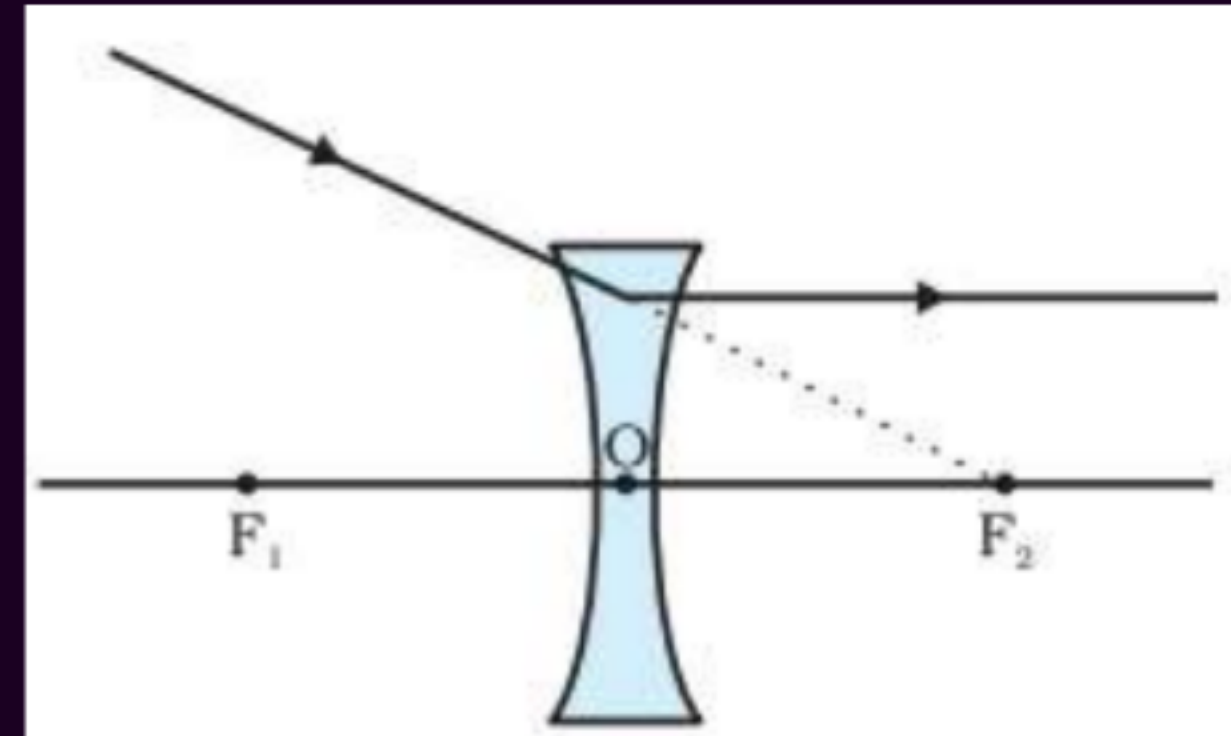
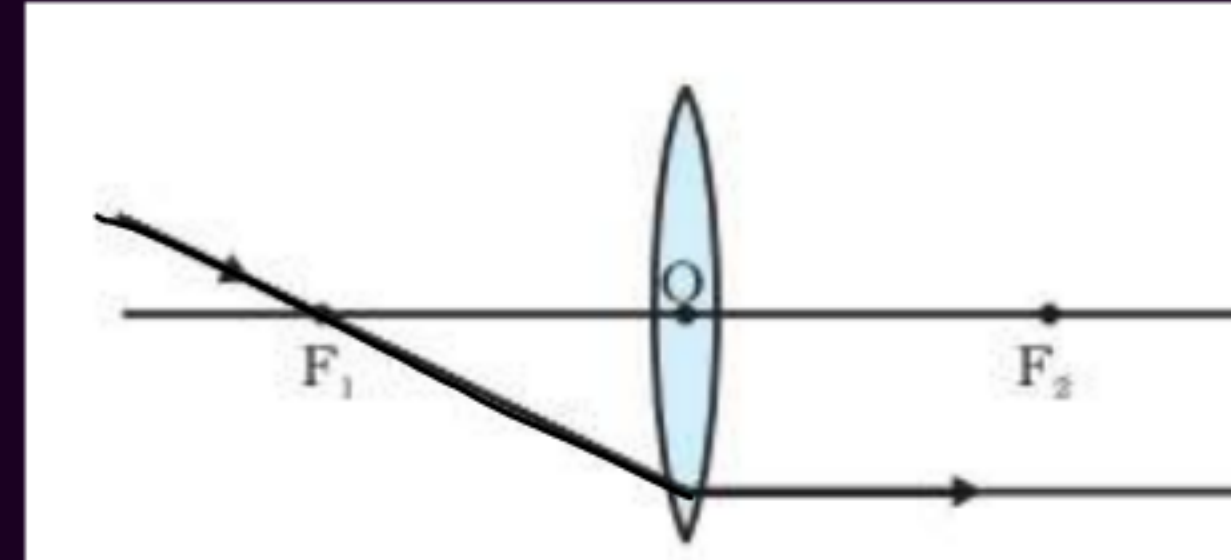
Rules to obtain Image: (i)

- **For Convex Lens:** A ray of light coming from the object and travelling parallel to the principal axis, after refraction through a convex lens, passes through the principal focus on the other side of the lens.
- **For Concave Lens:** A ray of light coming from the object and travelling parallel to the principal axis, after refraction through a concave lens, appears to diverge from the principal focus on the same side of the lens.



Rules to obtain Image: (ii)

- **For Convex Lens:** A ray of light passing through the principal focus of a convex lens, after refraction, emerges parallel to the principal axis.
- **For Concave Lens:** A ray of light which appears to meet at the principal focus of a concave lens, after refraction, emerges parallel to the principal axis.

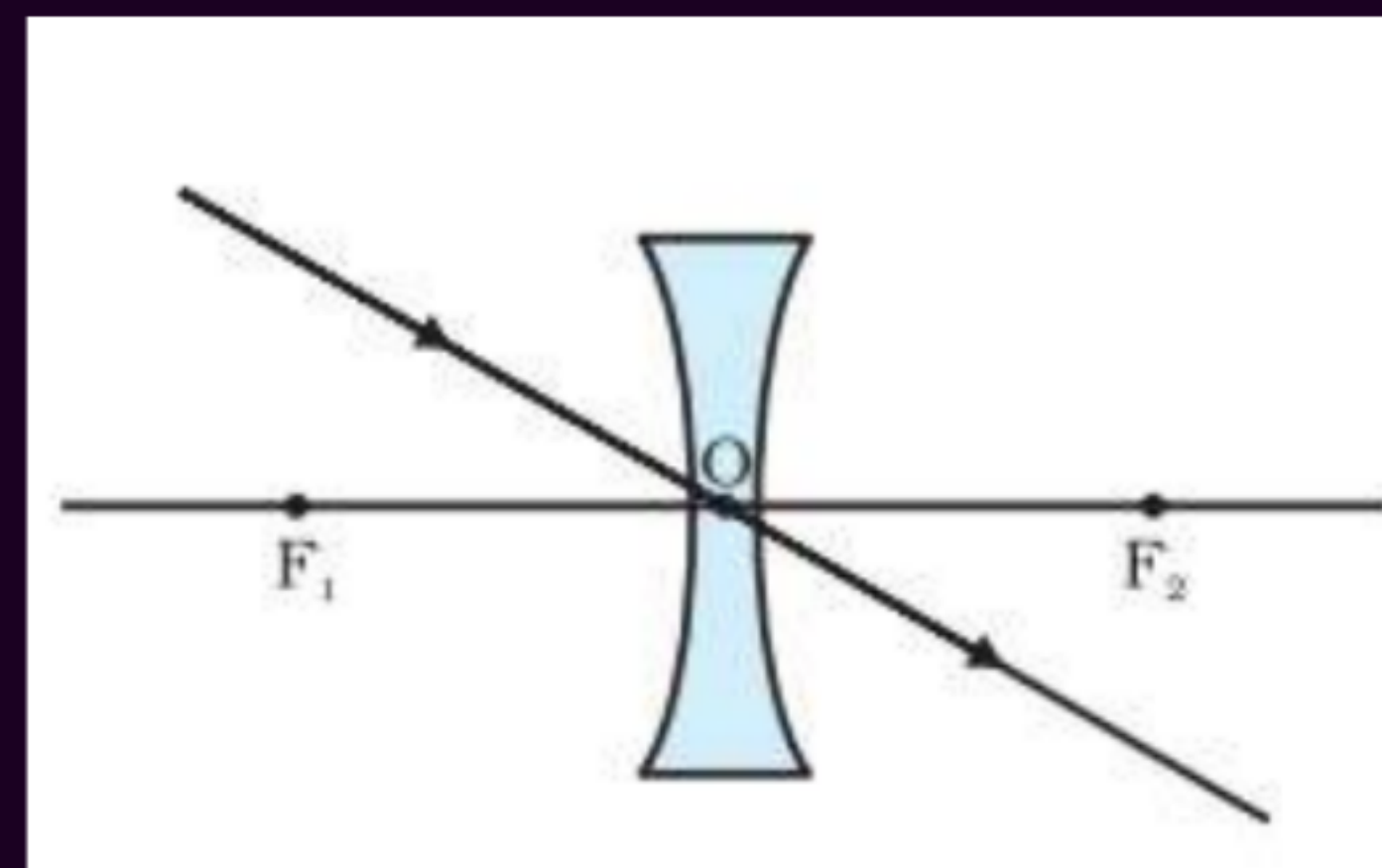
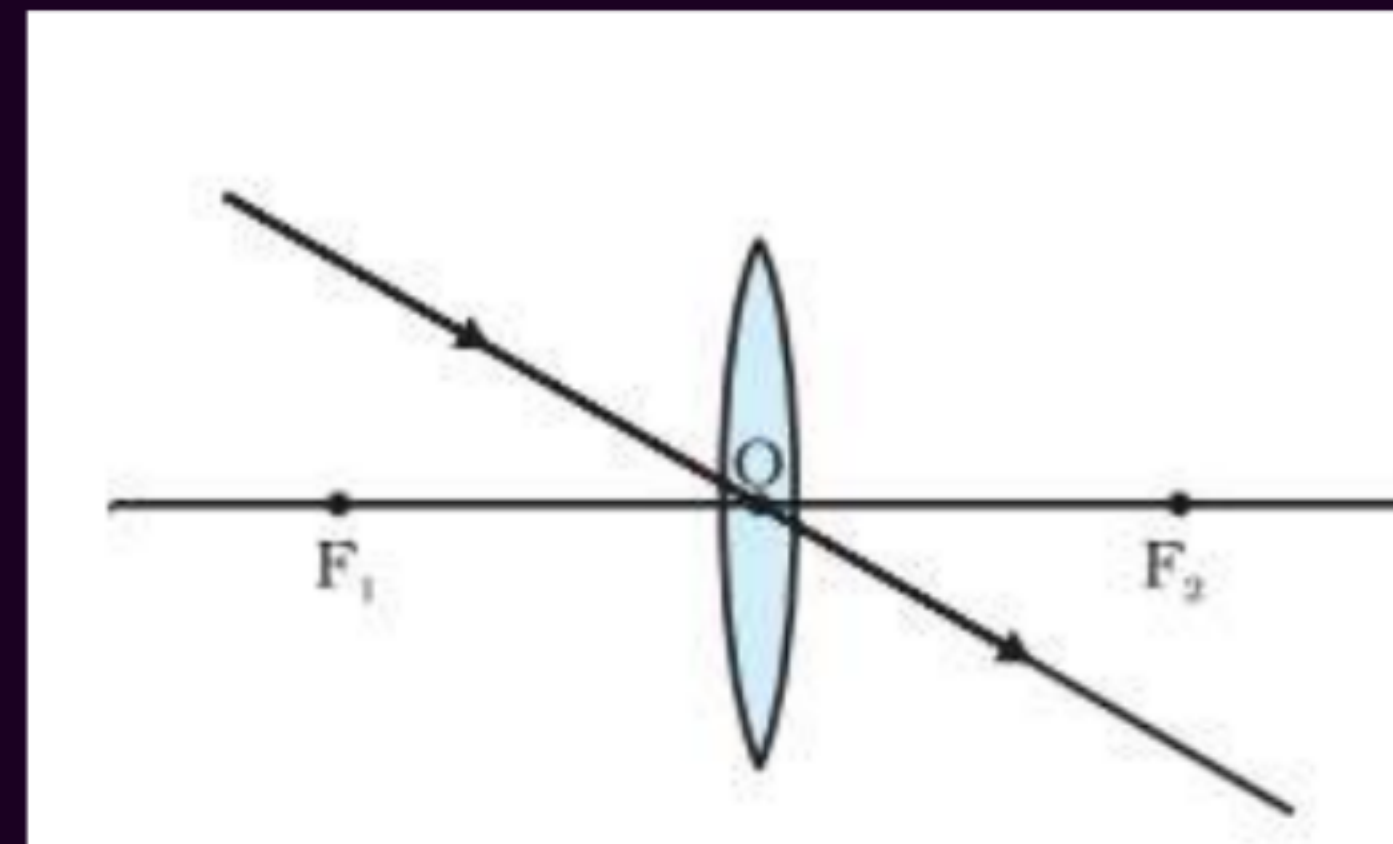


Rules to obtain Image: (iii)

A ray of light passing through the optical centre of a lens will emerge without any deviation.

This rule is true for both:

- *Convex lens*
- *Concave lens*

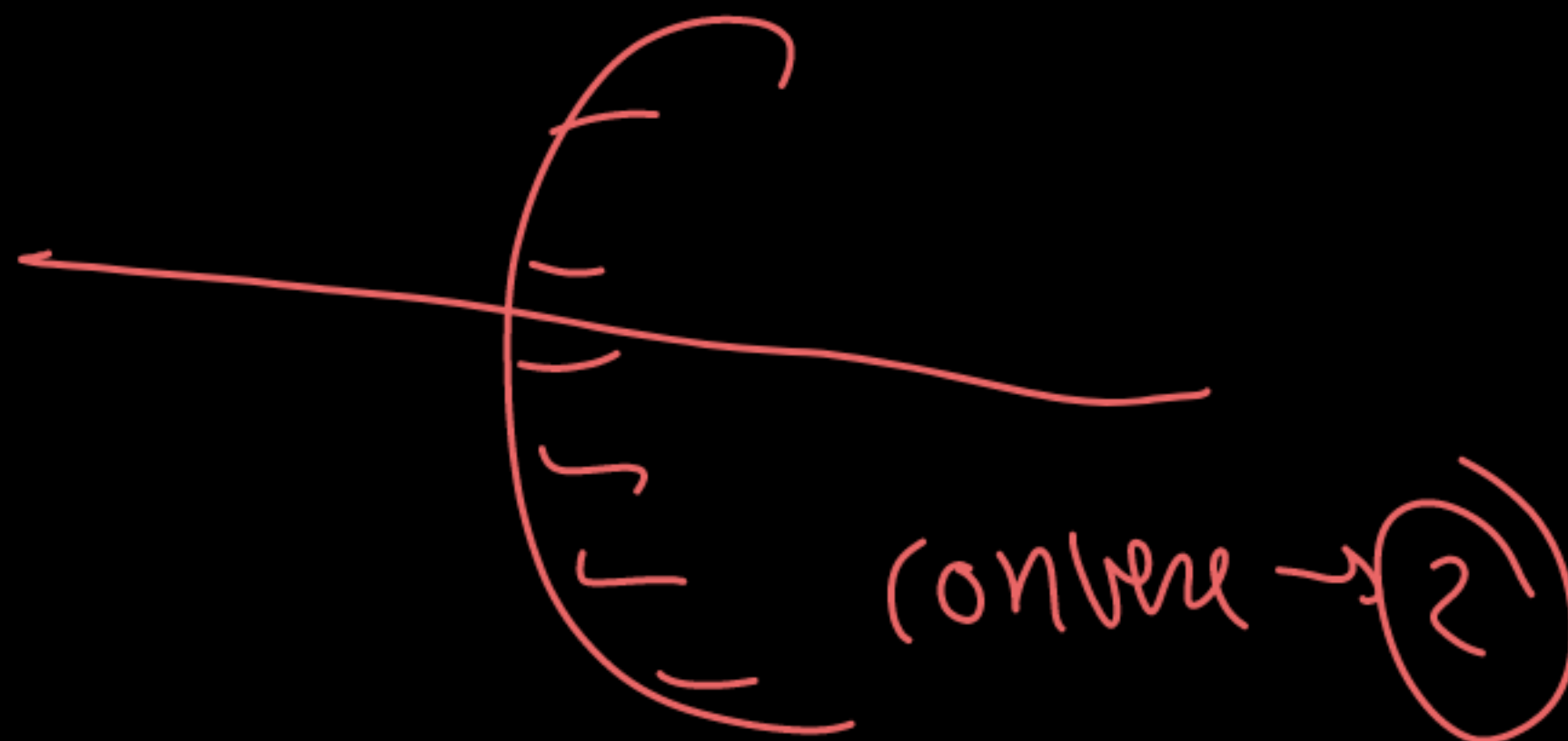




Concave $\rightarrow$ (6)



convex $\rightarrow$ 6 same



convex $\rightarrow$ (2)



convex
(2)

Image Formation: Convex Lens

1. Object at infinity

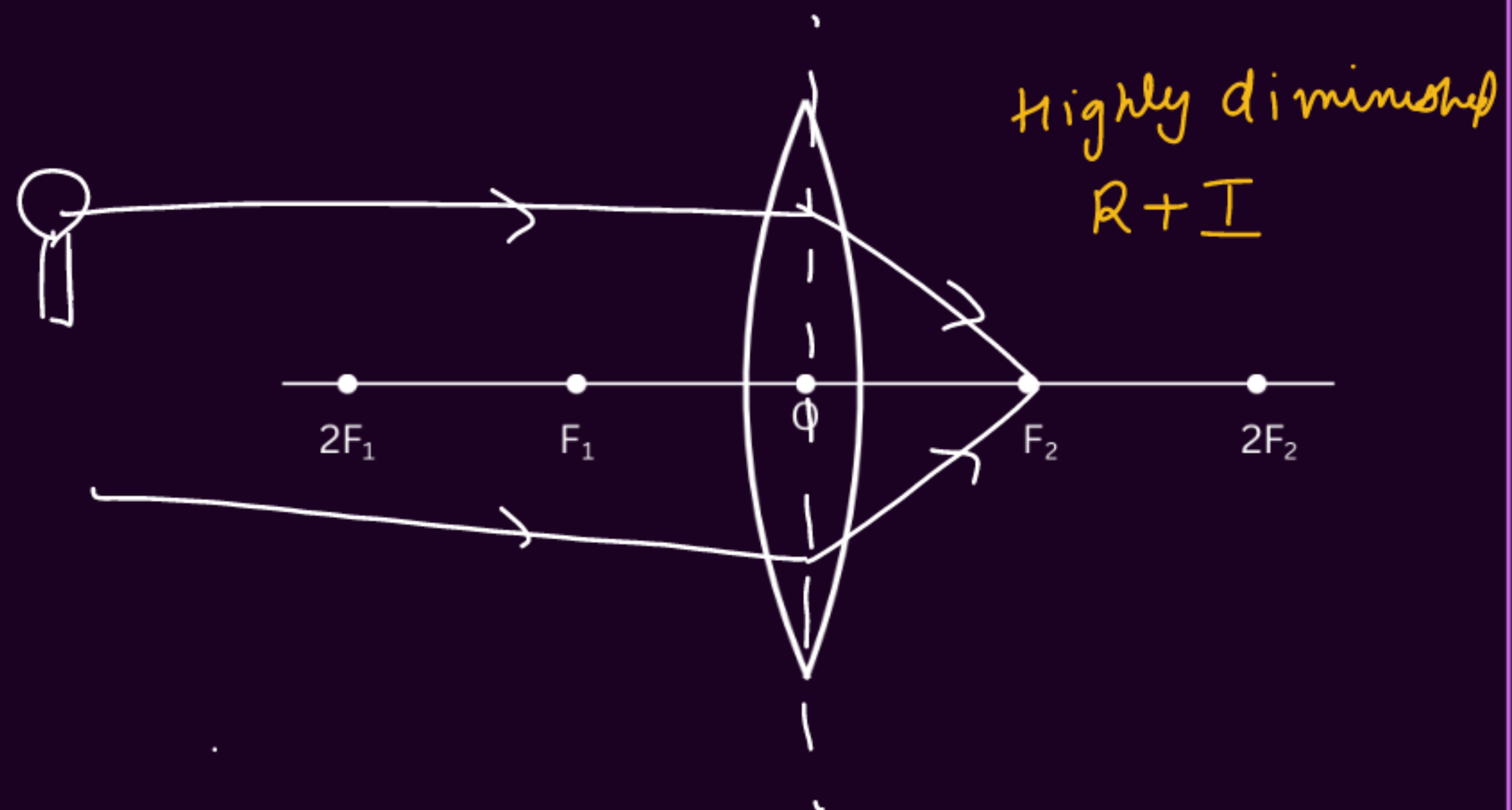


Image Formation: Convex Lens

2. Object beyond $2F_1$

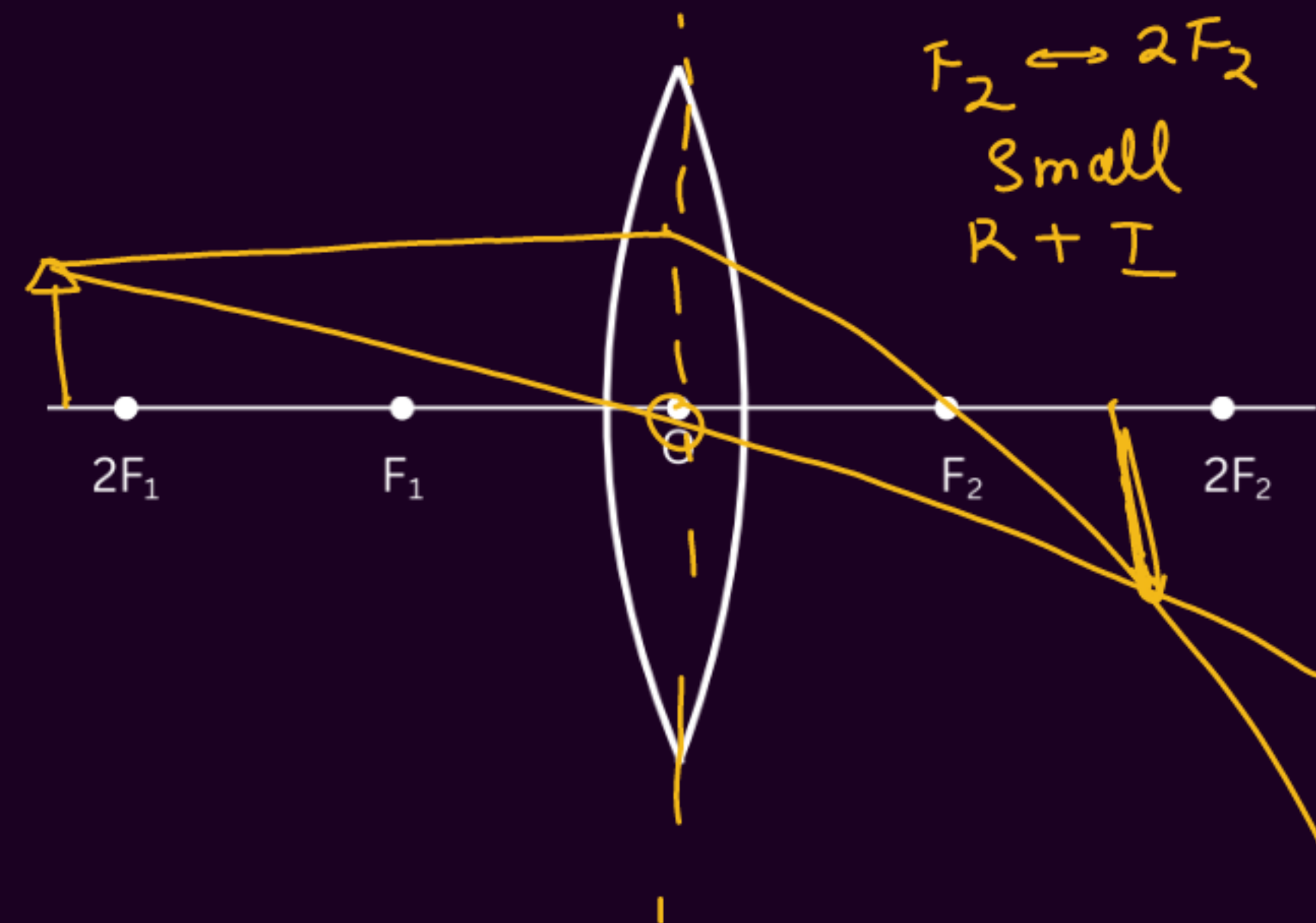


Image Formation: Convex Lens

3. Object at $2F_1$

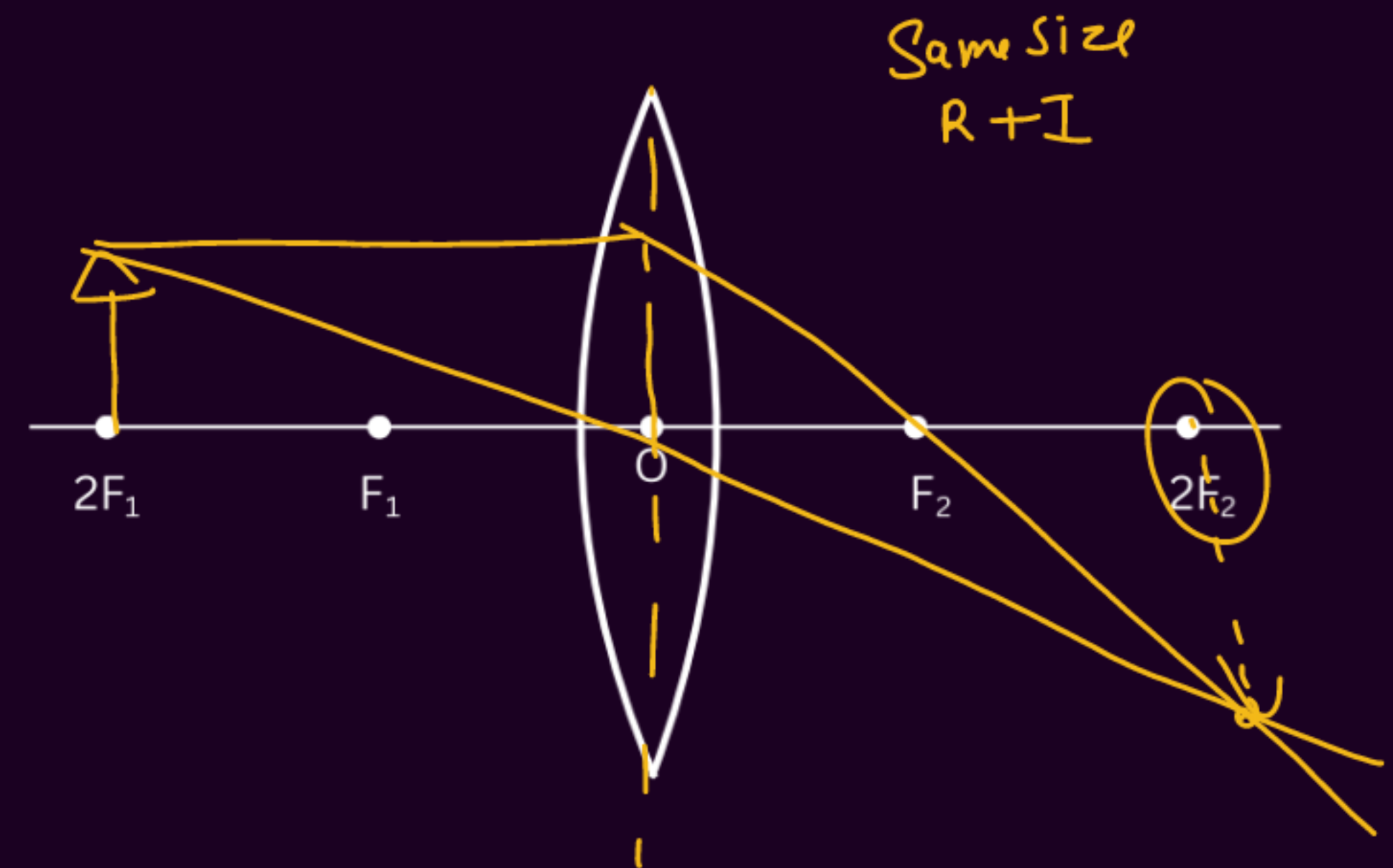


Image Formation: Convex Lens

4. Object between $2F_1$ and F_1

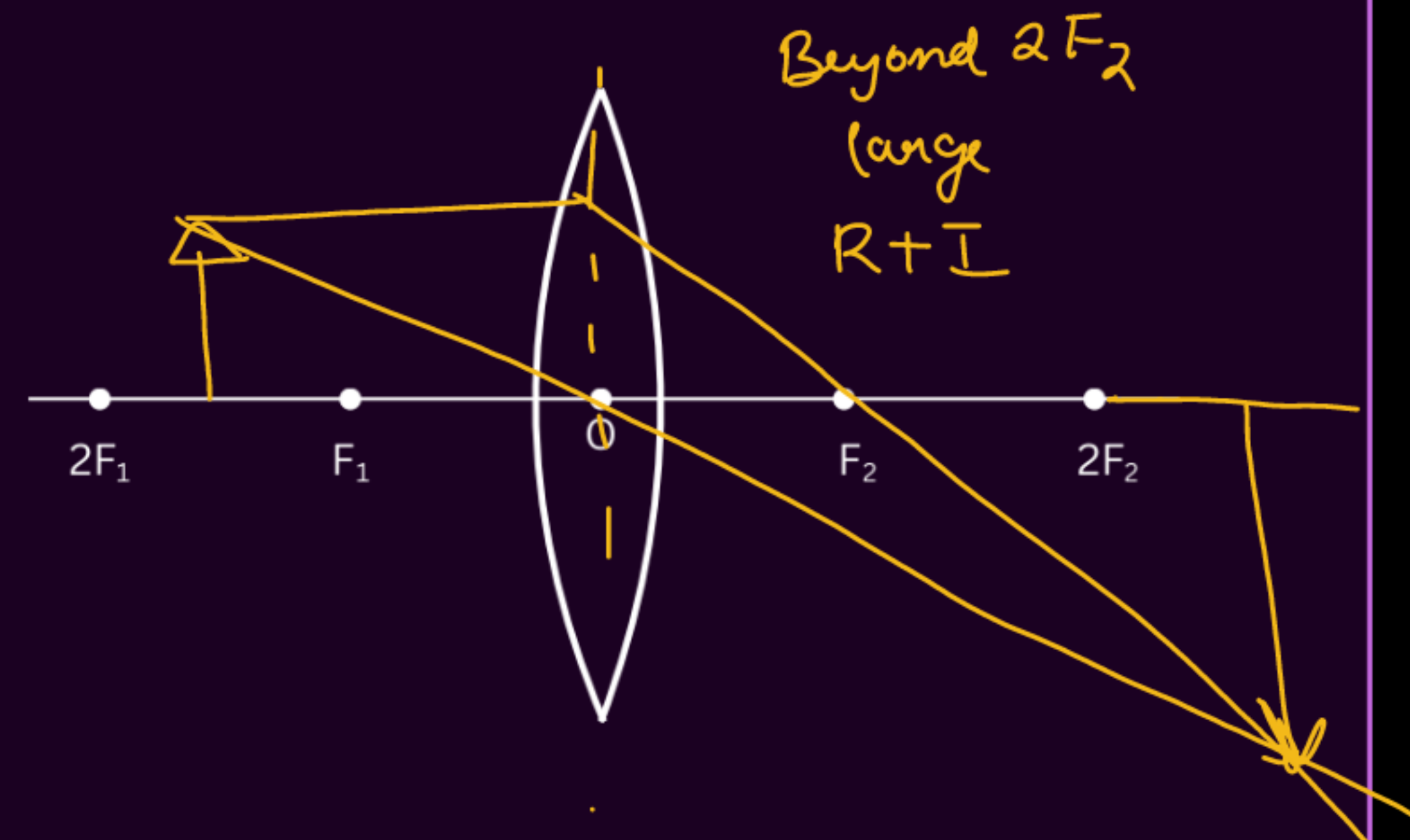


Image Formation: Convex Lens

5. Object at F_1

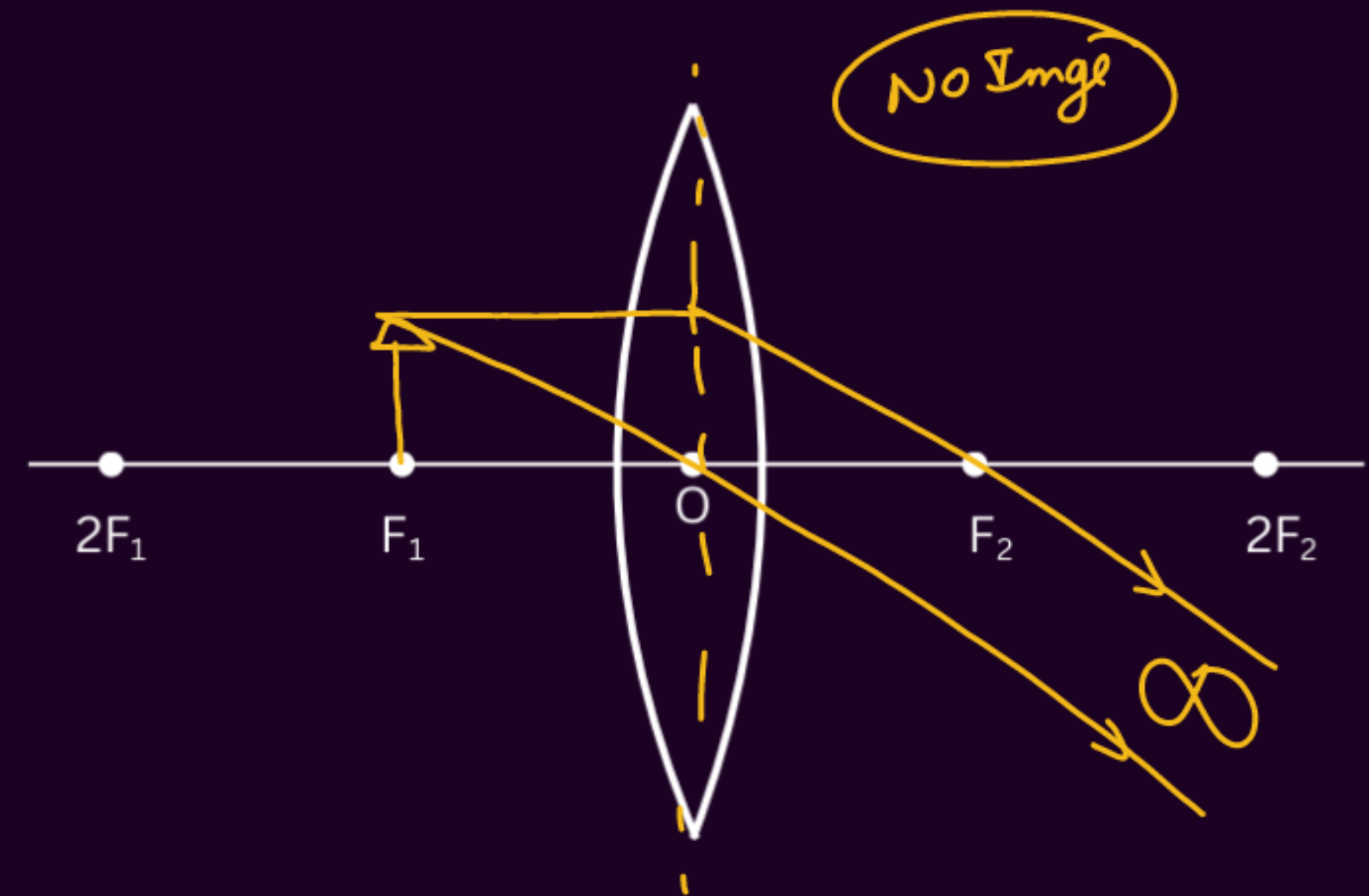
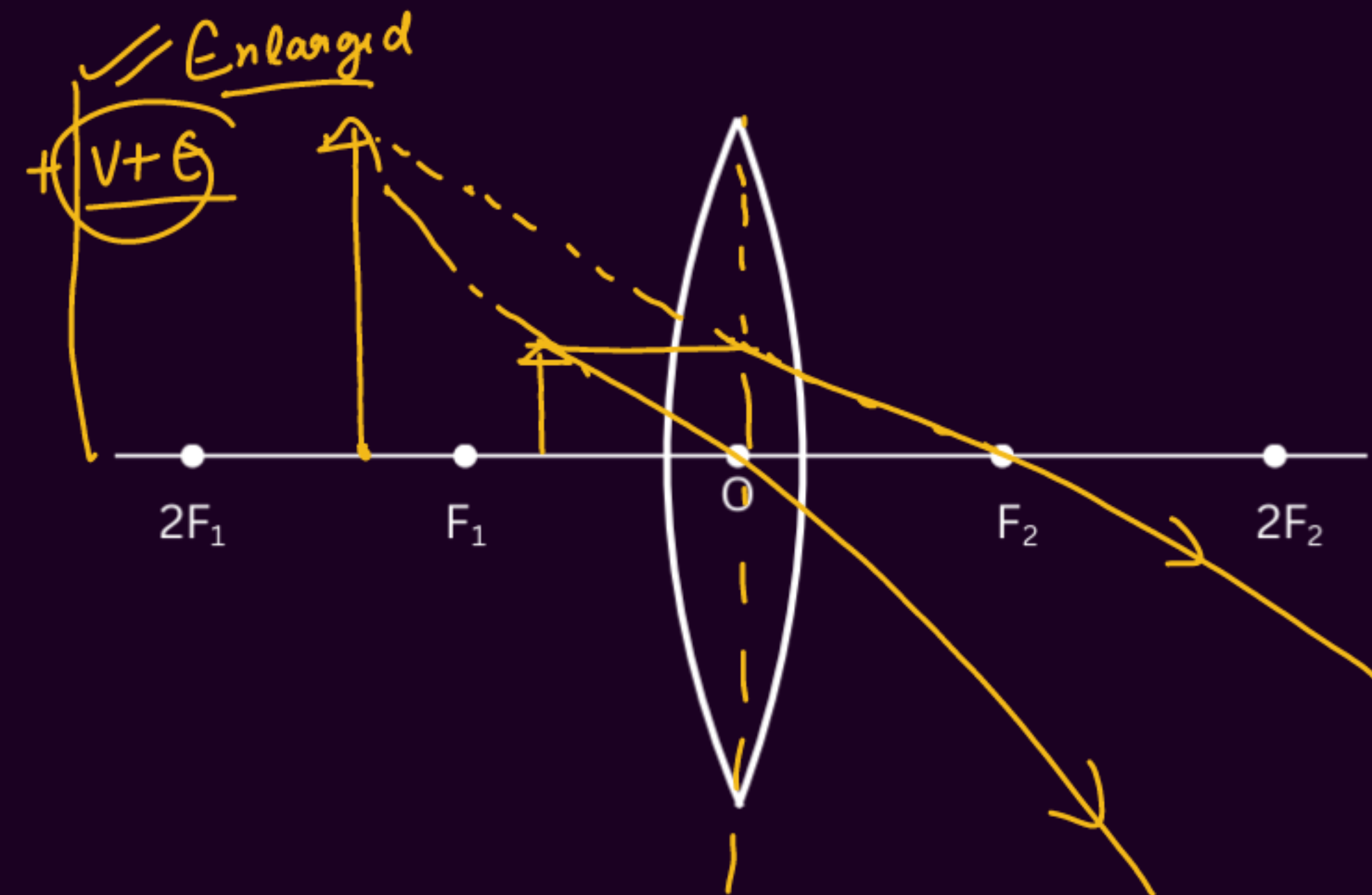


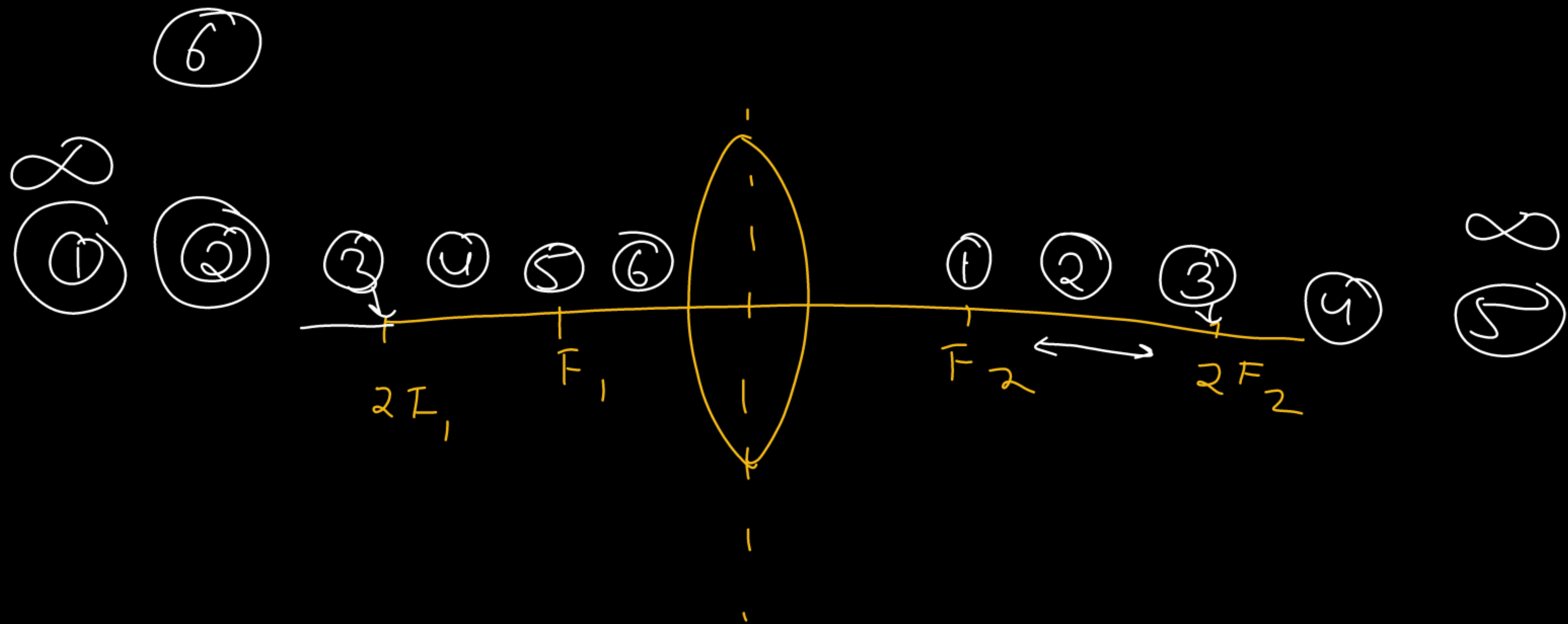
Image Formation: Convex Lens

6. Object between F_1 and O



Nature, position and relative size of the image: Convex lens

Position of the object	Position of the image	Relative size of the image	Nature of the image
At infinity	At focus F_2	Highly diminished, point-sized	Real and inverted
Beyond $2F_1$	Between F_2 and $2F_2$	Diminished	Real and inverted
At $2F_1$	At $2F_2$	Same size	Real and inverted
Between F_1 and $2F_1$	Beyond $2F_2$	Enlarged	Real and inverted
At focus F_1	At infinity	Image would not be formed	
Between focus F_1 and optical centre O	On the same side of the lens as the object	Enlarged	Virtual and erect



Uses of Lens

Convex Lens

- **Magnifying Glass** – Convex lenses enlarge images, making them useful in magnifying glasses.
- **Eyeglasses for Hyperopia (Farsightedness)** – They help focus light correctly on the retina.
- **Cameras and Microscopes** – Used in focusing light to form clear, real images.



Image Formation: Concave Lens

1. Object at infinity

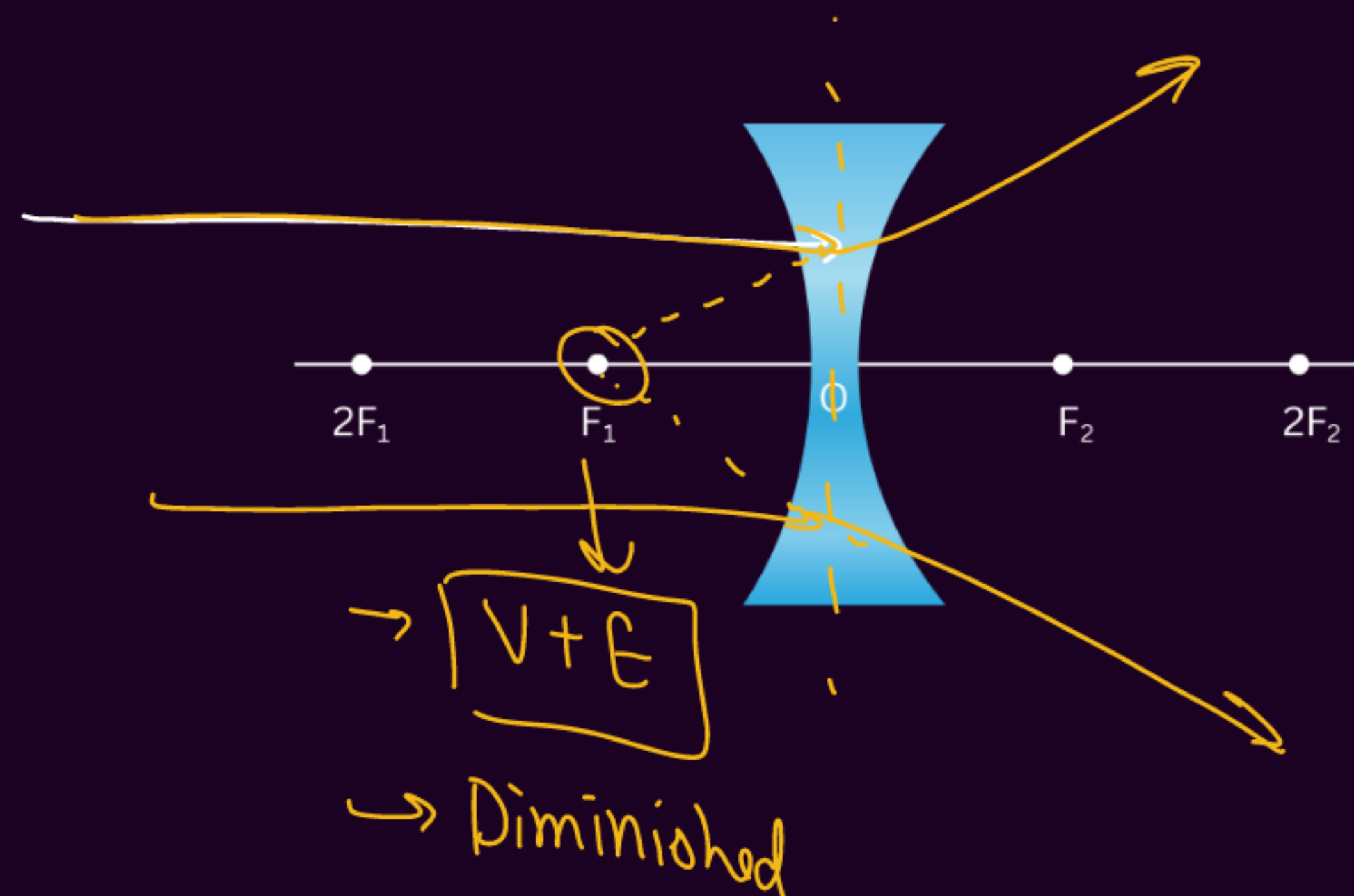
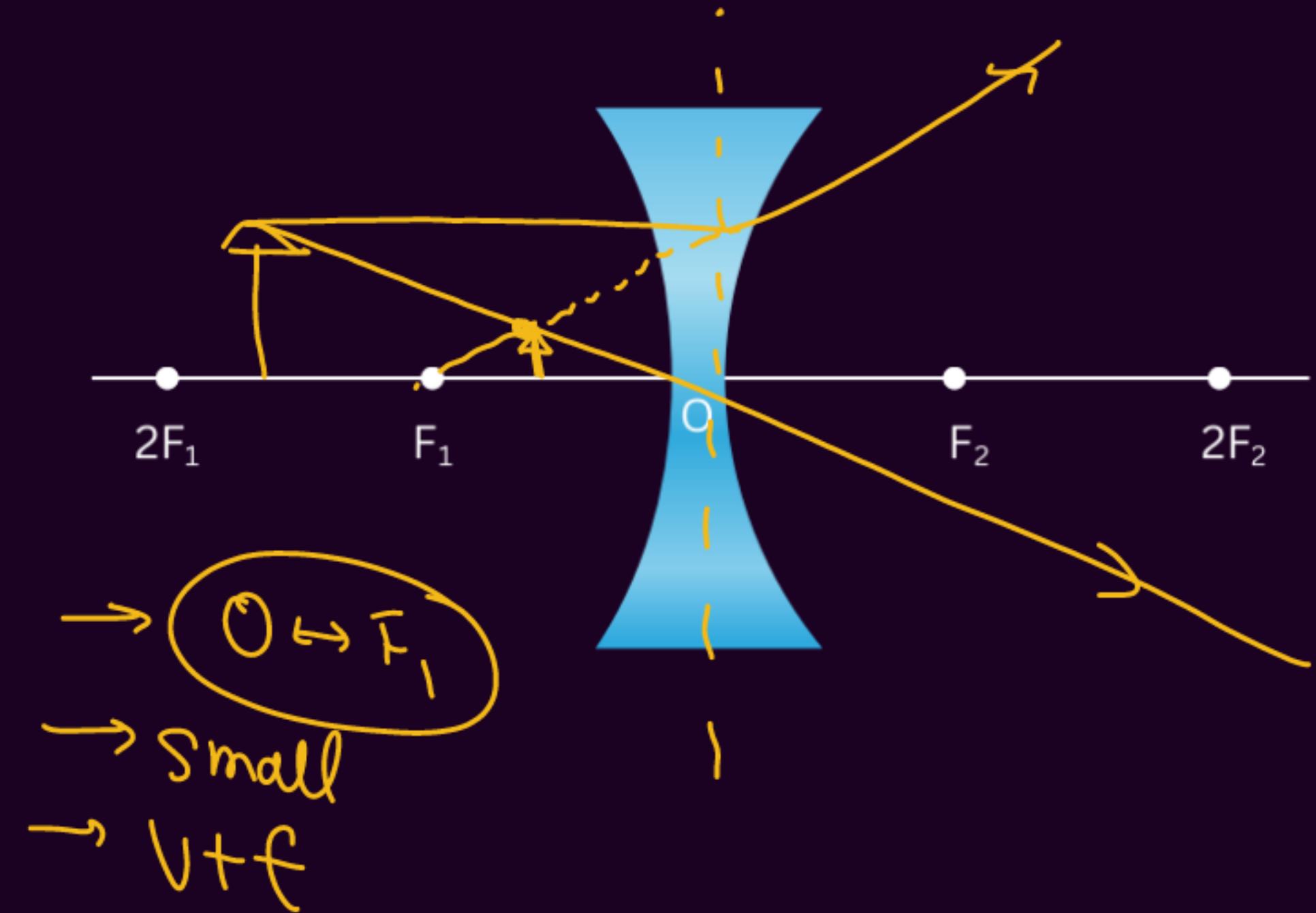


Image Formation: Concave Lens

2. Object between infinity and optical centre

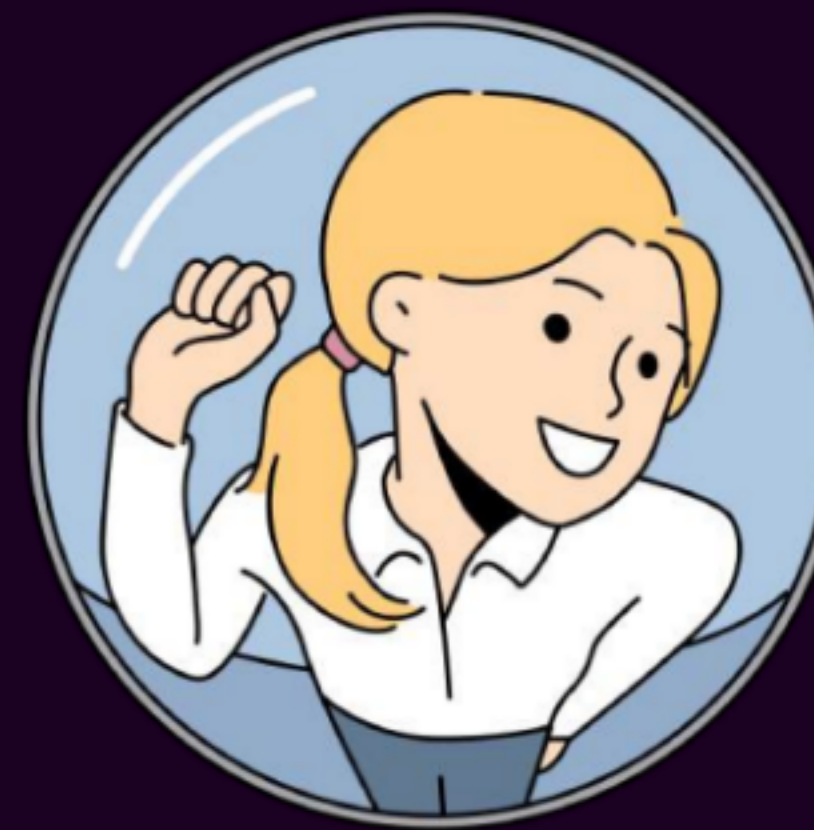


Nature, position and relative size of the image: Concave lens

Position of the object	Position of the image	Relative size of the image	Nature of the image
At infinity	At focus F_1	Highly diminished, point-sized	Virtual and erect
Between infinity and optical centre O of the lens	Between focus F_1 and optical centre O	Diminished	Virtual and erect



Uses of Lens



④ Concave Lens

- Eyeglasses for Myopia (Nearsightedness) – They help diverge light rays so they focus properly on the retina.
- ✓ Peepholes in Doors – Concave lenses provide a wider view of the outside area.
- ✓ Lasers and Flashlights – Help spread out light for better coverage.



Lens Formula and Magnification

$$\frac{1}{v} - \frac{1}{u} = \frac{1}{f}$$

This equation shows the link between the object distance (u), image distance (v), and the focal length (f) of the lens.

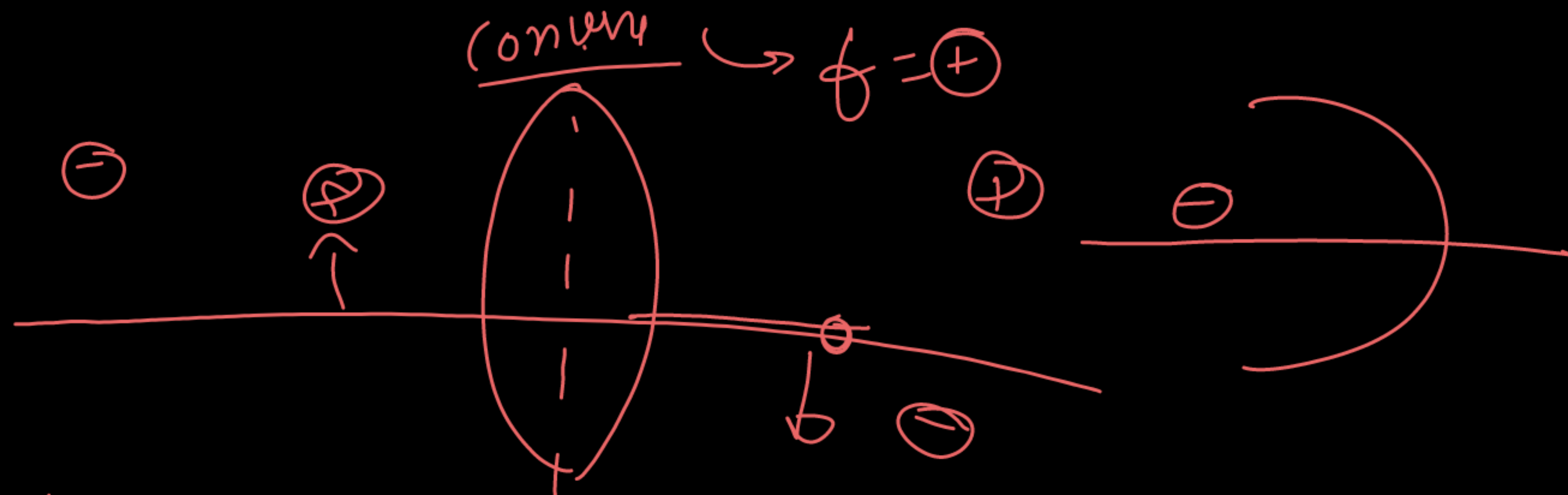
Magnification $\frac{1}{v} + \frac{1}{u} = \frac{1}{f}$

- A lens's magnification is the ratio of the image's height to the object's height, just like in spherical mirrors.
- It is represented by the m.

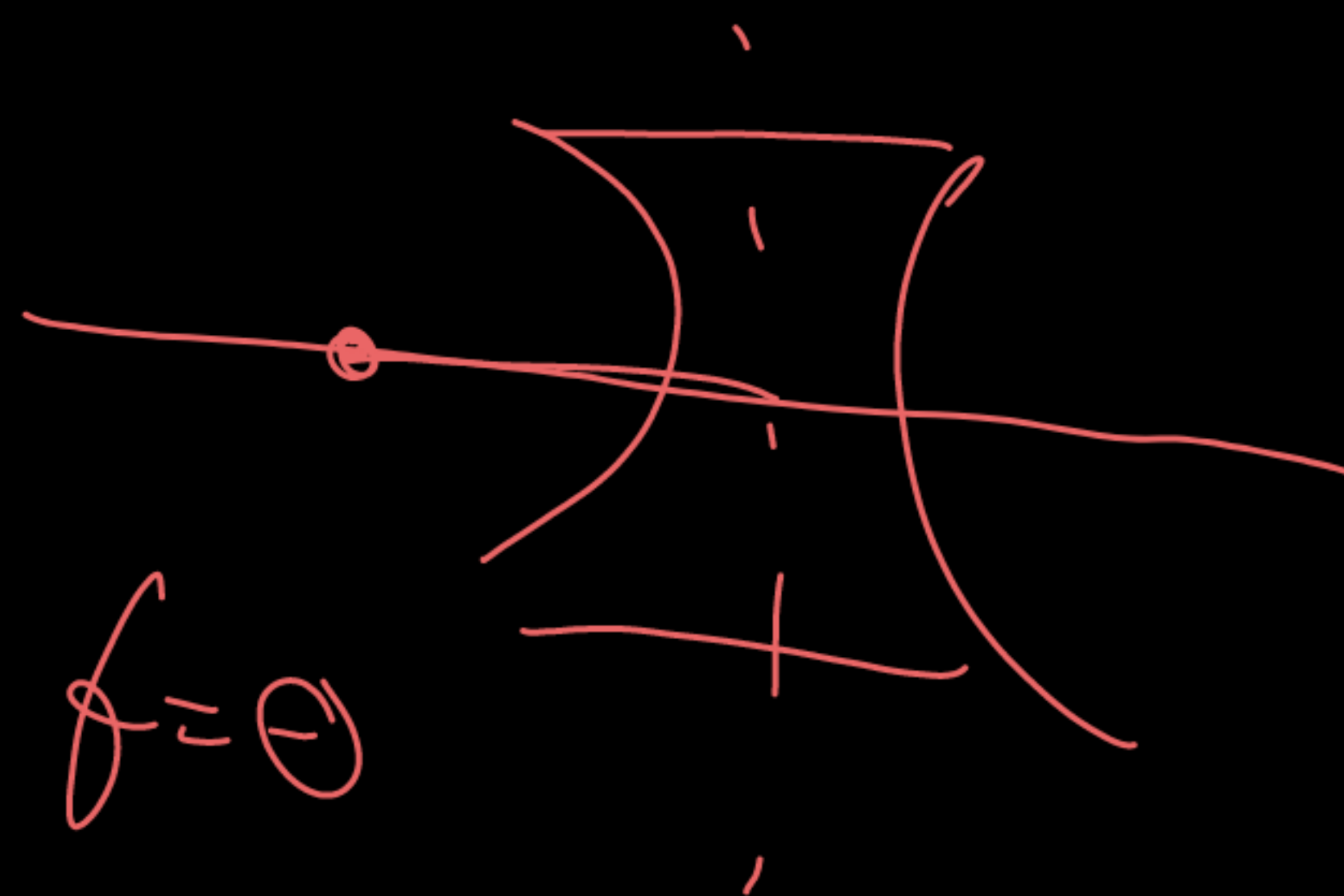
$$m = \frac{h_i}{h_o} = +\frac{v}{u}$$

$$m = \frac{v}{u}$$

$$m = \frac{\text{Height of the Image}}{\text{Height of the object}} = \frac{h'}{h}$$



$$u = \ominus$$



Power of the Lens

$$P = \frac{1}{f(\text{m})} = 1\text{D}$$

- The convergence and divergence of the parallel incident light on the surface of the lens.
- The power of a lens is defined as the reciprocal of its focal length. It is represented by the letter P .

SI Unit of Power of a lens is Diopetre (D).

1 diopetre is the power of a lens whose focal length is 1 metre [1D = 1m⁻¹].

Power of a convex lens is *positive* and that of a concave lens is *negative*.

$$\text{power of a lens (D)} = \frac{1}{\text{focal length (m)}}$$

$$P = \frac{1}{f}$$



$$f = -$$

$$P = -$$

$$P = \frac{1}{\text{focal length (m)}} = \frac{100}{f \text{ (cm)}}$$

'D'



convex

$$P \text{ of convex lens} = +$$

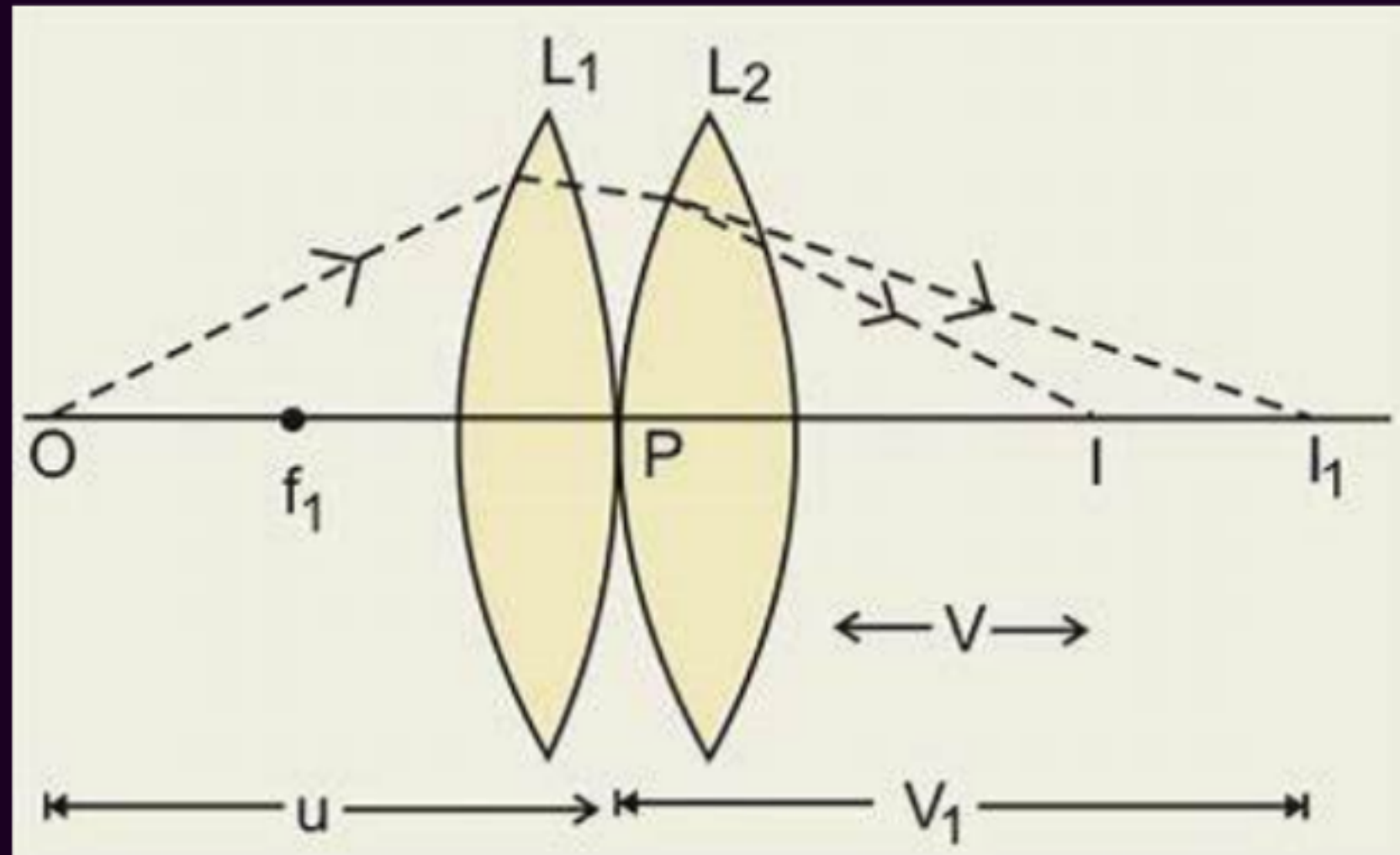
Combination of Lenses

- When multiple lenses are used together, their total power is added algebraically.

$$\text{Total Power: } P = P_1 + P_2 + P_3 + \dots$$

- This concept is key in designing instruments like glasses and microscopes.*

- If two lenses are placed in contact:**



$$\frac{1}{F} = \frac{1}{f_1} + \frac{1}{f_2}$$

OR

$$\text{Power} = P_1 + P_2$$